

CS201

Mid-Term

Quiz by Attiq Kundi

SUBSCRIBE "Knowledge For Us" ON YOU TUBE**100 % CORRECT MCSQ****AND Many MoRe****CS 201****Also Added Today Quiz****15-june-2022**

The ASCII code of null character is -----.

(000) correct answer

Which step is not performed during a proper program design?

(Write details about the performance of program) correct answer

How many elements are stored in the following ? int matrix [4][5];

(20) correct answer

Which of the following is not a valid variable name in c++?

(@studentName) correct answer

If an array has 50 elements. What is allowable range of subscripts?

(0-49) correct answer

<, <=, >, >= are called -----operators.

(Relational) correct answer

Why we use "cin"?

(To read data from keyboard) correct answer

!(x<3) means in c++ that

(x is greater than 3) correct answer

The data in the memory is -----

(volatile) correct answer

Each array declaration must provide the information about all these things except -----

The location of the first element to be stored

What will be the output of the program if x=10? If

(x!=0&& x<=10)

{

Cout <<"condition 1 is true";

}

If(x>=10 && x<=20)

{

Condition 1 is true condition 2 is true

C is widely known development language of _____ operating system.

Windows

A repetition statement specifies that an action is to be continued while some _____ remains true.

Condition

_____ is a substitute of multiple if statement.

If—elseif statement

If a _____ finds an error. It stops translating without generating any executable code.

Compiler

Function prototype is written .

Before call of that function

The _____ is an unconditional branch of execution .

Goto statement

What will be the output of the following code snippet?

String str1 = "Hello beautiful";

String str2 = "yes we can";

```
String str3 = strcpy(str1,str2);
Cout<<str3<<endl;
```

Hello beautiful Yes we can

In two dimensional array .We use _____,for oop(s)"to populate the array.

2

eof is used to check for the _____ of file when a file is being read.

End

How many times the following loop will execute? Int j

= 3;

While(j>0)

{

Cout << "statements" << endl;

J=2;

}

2

Null character is used to indicate the _____ of string.

End

There is a pointer named ptr of type float. Which type of variable address can be stored by ptr?

Float

How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed?

```
intPtr +=2;
```

4

We must include the header file _____ to convert the value of one type into another type using built-in functions.

iostream.h

Use of word processor for writing the code of a program is not good because.

It has no compiler

_____ Loop executes atleast one times.

Do while

The data . contained in the structure. is defined in the _____

Curly braces

What is the use of ios::trunk mode?

To truncate an existing file to zero

CS201 Solved Quiz No-1 December 13, 2015

We use _____ to open a file or stream for extraction

in

Null character is represented by _____ in C++.

/0

An object of _____ class can be created for saving the current position of any opened file before reading from the file.

Streampos (pg215)

_____ Returns true if c is a digit or a letter and false otherwise.

int isalnum(int c)

We use _____ to open a file or stream for insertion.

out

Base address is the memory address of _____ element of an array.

1st

_____ allow us to have a memory location and use it as int or char interchangeably

union

To read command-line arguments, the main() function itself must be given _____ arguments.

2

Individual characters in a string stored in an array can be accessed directly using array. _____.

Subscript

To manipulate n-dimensional array _____ nested loops are required.

n-1

A character is stored in the memory in _____

String

What will be the size of following array?

```
int arr[29];
```

29

An object of _____ class can be created for saving the current position of any opened file before reading from the file.

Streampos

We use _____ to open a file or stream for extraction.

In

We use _____ to open a file or stream for insertion.

Out

_____ are conventional names of the command line parameters of the „main()“ function.

**__argc‘ and
__argv‘**

In _____, we have different data members and all of these have their own memory space.

Structures

How we declare an array of integer data whose size is 20 with array name “count”.

**int
count[20];**

A _____ is an array of characters that can store number of character specified.

String

File. seekg (-10L, ios:cur)

backward

direction

What will be the output of the following while loop? `int counter = 0 ; while(counter < 15) cout << counter << " "; counter++;`

Compile time error

We can access a global variable _____

From the functions only

From anywhere in the program

A repetition statement specifies that an action is to be continued while some _____ remains true.

Condition

Loops are _____ Structure.

Repetition

_____ operator is used to pass the address of a variable in call by reference method.

&

How many times the following do-while loop will execute? `int k = 10; do { cout << "Statements" << endl; k -= 2; } while(k>0);`

5

What is the output of the following code if the 3rd case is true `switch (var) { case „a“: cout<<„apple“<<endl; case „b“: cout<<„banana“<<endl; case „m“: cout<<„mango“<<endl; default: cout<<„any fruit“<<endl; }`

mango any

fruit

CS201 SOLVED QUIZ NO 1 DECEMBER 09, 2019

The ASCII code of null character is_____

010

Each array declaration must provide the information about all these things except_____

The location of the first element to be stored

A record is a group of related_____

Data

Pointers store the_____

Value of the variable

We can say that a pointer is a variable that holds the_____ address.

Source code

afseek(-10, SEEK_CUR,

Backward direction

Deferencing operator is represented by_____

***, +, -, non of the given**

A precise sequence of steps to solve a problem is called

Program

The Compiler of C language is written in

C Language

Initialization of variable at the time of definition is,

Good Programming

In if structure the block of statements is executed only,

When the condition is true

Which of the following function(s) is/are included in **stdlib.h** header file?

double atof(const char *nptr)
int atoi(const char *nptr)

1 and 2 only

Dealing with structures and functions passing by reference is the most economical method

True

Pointer is a variable which store,

Memory Address

Preprocessor program perform its function before _____ phase takes place.

Compiling

Which of the following can not be a variable name?

10area

Which looping process is best, when the number of iterations is known?

while

Which character is inserted at the end of string to indicate the end of string?

null

How many bytes are occupied by declaring following array of characters?

char str[] = —programming!;

12

11 plus one for null char (11+1= 12)

Which of the following header file defines the **rand()** function?

stdlib.h

Commenting the code _____

Makes a program easy to understand for others.

What's wrong with this for loop? for (int k = 2, k <=12, k++)

the commas should be semicolons

For which array, the size of the array should be one more than the number of elements in an array?

char

To Which category of the software “Compiler and Interpreter” belongs?

They belong to system software.

What is the result of the expression $x = 2 + 3 * 4 - 4 / 2$

12

Write a declaration statement for an array of 10 elements of type float. Include an initialization statement of the first four elements to 1.0, 2.0, 3.0 and 4.0.

float tmp [10] = {1.0,2.0,3.0,4.0};

Write down the output of the following code? `int array[7], sum = 0;`
`for(int i=0;i<7;i++)`
`{`
`array[i] = i; sum+= array[i];`
`}`
`cout<< “ Sum = “ <<sum;`

21

what will be the output of the following segment of C++ code?

```
int A[5] = {1 , 2, 3, 4};
int i;
for (i=0; i<5; i++)
{
A[i] = 2*A[i];
cout << A[i] << " ";
}
```

2 4 6 8 0

Write a C++ program that will determine if a departmental store customer has exceeded the credit limit on a charge account.

Program should input the following facts in five variables

1. Account number
2. Balance at the beginning of month (Beginning balance)
3. total of all items charged by customer this month (charges)

4. total of all credits (credits)
5. allowed credit limit Calculate the new balance

New balance = Beginning balance + charges – credits

Determine if **new balance** exceeds the **allowed credit limit**. For those customers whose credit limit is exceeded. The program should display the message “Credit Limit exceeded.”

by Attia Kundi

What is the output of the following statement?

```
int i = 2.5; do { cout << i * 2; } while (i > 3 && i < 10);
```

error

What does !(7) evaluate to in decimal where ! is a NOT operator?

7

The condition in while loop may contain logical expression but not relational expression.

False

Searching is easier when an array is already sorted

True

If an array has 100 elements, what is allowable range of subscripts?

- 99

What will be the value of „a“ and „b“ after executing the following statements?

```
a = 3;
```

```
b = a++;
```

4, 3

What will be the correct syntax to initialize all elements of two-dimensional array to value 0?

```
int arr[2][3] = {{0},{0}} ;
```

Which of the following function returns the size of a string variable?

s strlen()

What will be the range of numbers generated by function rand () % 9?

- (a) 0 to 9
- (b) 1 to 9
- (c) 0 to 8
- (d) 1 to 8

Computer can understand only machine language code.

(e) True

What does $5 \wedge 6$, evaluate to in decimal where „ $\wedge$ “ is Exclusive OR operator?

False

It mean

5 = 0101

6 = 0110

$5 \wedge 6 = 0011$

If both input is same then the output is 0 and if different then output is 1

If the file is not properly closed in the program, the program_____.

Terminate normally

Which of the following header file include string conversion functions?

string.h

In Program commenting the code liberally is

It need to be self-contained and understandable. Comments should be placed liberally. The comments should explain the logic, not the mechanics. Try to avoid fancy programming.

Which header file must be included while handling files?

<fstream.h>

What is meant by C++ statement: `const int *ptr = &x;`

ptr is a pointer to data of type const int type. And to assign the address of x to pointer ptr

What is a truth Table?

Solution:-

We know the concept of truth table. The truth tables are very important. These are still a tool available for analyzing logical expressions. We will read logic design in future, which is actually to do with chips and gate. We find it difficult to evaluate a complicated logical expression. Sometimes the logic becomes extremely complicated so that even writing it as a simple syntax statement in any language.

An array day is declared as: `int day[] = {1, 2, 3, 4, 5, 6, 7};`

How many elements does array 'day' has?

Solution:-

7 elements

If the declaration is changed as: `int day[7] = {1, 2, 3, 4, 5, 6, 7};` How many elements does array 'day' has?

Solution:-

7 elements

What are similarities and differences between Structures and Unions?

In structures, we have different data members and all of these have their own memory space. In union, the memory location is same while the first data member is one name for that memory location. However, the 2nd data member is another name for the same location and so on. Consider the above union (i.e. `intOrChar`) that contains an integer and a character as data members. What will be the size of this union? The answer is the very simple. The union will be allocated the memory equal to that of the largest size data member. If the `int` occupies four bytes on our system and `char` occupies one byte, the union `intOrChar` will occupy four bytes

Write a **void function()**; that takes integer numbers from the user and then displays the sum of odd and even numbers entered by the user. Your program should terminate if user enters a negative number

Solution:-

```
#include<iostream.h>
#include<conio.h>
void
function(void);
main()
{
function();
getche();
}
void function(void)
{
int n[5];
```

```

for(int i=0; i<=5; i++)
{
    cout <<"Enter Element = ";
    cin >>n[i];
}
cout <<"\nODD Inputs are = \n";
for(int i=0; i<=5; i++)
{

    if(n[i]%2==0)
    {
        cout <<n[i]<<endl;
    }

}
cout <<"\nEVEN Inputs are = \n";
for(int i=0; i<=5; i++)
{
    if(n[i]%2==1)
    {
        cout <<n[i]<<endl;
    }

}
}

```



The function of cin is

To read data from keyboard

In C/C++ language the header file which is used to perform useful task and manipulation of character data is

ctype.h

How many parameter(s) function getline() takes?

3

Word processor is

Application software

For which values of the integer _value will the following code becomes an infinite loop?

```
int
number
=1;
while
(true) {
    cout << number;
    if (number == 3) break;
    number += integer_value; }
```

only 0

Each pass through a loop is called a/an

Iteration

A continue statement causes execution to skip to

the statements following the continue statement

What is the correct syntax to declare an array of size 10 of int data type?

int name[10] ;

Consider the following code segment. What will the following code segment display?

```
int main(){
int age[10]
= {0}; cout
<< age ;
```

}

Value of first element of array

What will be the correct syntax to initialize all elements of two-dimensional array to value 0?

int arr[2][3] = {0,0} ;

How many bytes will the pointer *intPtr* of type int move in the following statement?
`intPtr += 3 ;`

12 bytes

If there are $2^{(n+1)}$ elements in an array then what would be the number of iterations required to search a number using binary search algorithm?

(n+1) elements

Which of the following operator is used to access the value of variable pointed to by a pointer?

*** operator**

The ____ statement interrupts the flow of control.

break

Analysis is the -----step in designing a program

First

Paying attention to detail in designing a program is _____

Necessary

Which programming tool is helpful in tracing the logical errors?

Debugger tool is helpful in tracing the logical errors.

Give the syntax of opening file „myFile.txt“ with „app“ mode using ofstream variable „out“.

```
ofstream outfile;
```

```
outfile.open ("myFile.txt "); // Open the file
```

What is the difference between switch statement and if statement.

In switch statement only one variable can be tested on various condition but using if we can tested multi variables in single statement.

Identify the errors in the following code segment and give the reason of errors.

```
ma
in(
){
int
x =
10
const int *ptr = &x ;
*ptr = 5 ;
}
```

```
main()
{
int x = 10;
const int *ptr
= &x ; cout
<<ptr;
}
```

If int array[10]; is an integer array then write the statements which will store values at Fifth and Ninth location of this array,

```
cout <<"Enter fifth
postion "; cin >>
array[4];
cout <<"Enter Ninth
postion "; cin >>
```

array[8];

Write a function BatsmanAvg which calculate the average of a player (Batsman), Call this function in main program (Function). Take the input of Total Runs made and Total number of matches played from the user in main function

When the if statement consists more than one statement then enclosing these statement in curly braces is,

Must

The while loop becomes infinite,

When the condition is always true

Which of the following function(s) is/are included in **stdlib.h** header file?

1 and 2 only

If we want to store a string "abc" in an array str then the size of this array must be at least,

2

No executable code will be generated if error is found during translation of the program using interpreter.

True

Word processor is

Application software

Which of the following is the correct syntax to print multiple values or variables in a single command using cout?

cout << "H" << x << "\n";

Which of the following is correct way to initialize a variable x of int type with

value 10?

int x = 10 ;

If there is more than one statement in the block of a for loop, which of the following must be placed at the beginning and the ending of the loop block?

braces { }

Name of an array is a constant pointer.

True

How many bytes will the pointer *intPtr* of type **int** move in the following statement?

intPtr += 3 ;

12 bytes

What will be the value of 'a' and 'b' after executing the following statements?

a = 3;
b = a++;

3, 4

Loader loads the executable code from hard disk to main memory.

True

Which of the following is used with bit manipulation?

Signed double

Which of the following values C++ use to represent true and false?

1 and 0

The argument of the isdigit() function is _____

character,

Which data type should be used to store the value 50.7 ?

Float

Why should goto statement be avoided in C/C++?

When structured programming was started, it was urged not to use the *goto* statement. Though *goto* is there in C language but we will not use it in our programs. It will adopt the structured approach. All of our programs will consist of sequences, decisions and loop. Because loop provide best platform to manipulate the data.

What operator do you use to assign a pointer the address of another variable or constant?

& sige

i.e. `int i;`

`int *`

`ptri; ptri`
`= &i;`

If there are 2^n elements in an array then what would be the number of iterations required to search a number using binary search and linear search? Marks: 3

Convert the following switch statement into if statements. Marks: 5

```
switch
(operator) {
case '+':
    result = op1 + op2;
    break;
case '-':
    result = op1 - op2;
    break;
case 'x':
case '*':
    result = op1 * op2;
```

```
break;  
case '/':  
    result = operand1 / operand2;  
    break;  
default:  
    cout << "Unknown operator" ;  
}
```

```
if(operator==' +')
```

```
{
```

```
    result = op1 + op2;
```

```
}
```

```
else if(operator==' -'  
-')
```

```
{
```

```
    result = op1 - op2;
```

```
}
```

```
els  
e
```

```

if (operator=='*')
    {
        result = op1 * op2;
    }

else
if (operator=='/')
    {
        result = op1 / op2;
    }

else
{
cout << "Unknown operator" ;

}

```

Q22

Write a recursive function that takes character array and starting subscript as arguments. In each recursive call, the function should display the string from subscript to the end of string. The starting subscript in first call should be 0. In each successive call, the subscript should increase by one and function should print the array from subscript to the end of string. The function should stop processing and return when null character encounters.

Suppose the char string passed to the function is, "SampleString", then the function will print output as follows,

Sample
String
ampleSt
ring
mpleStri
ng
pleStrin
g
so on....

Marks: 10

A precise sequence of steps to solve a problem is called

Program

The Compiler of C language is written in

C Language

Initialization of variable at the time of definition is,

Good Programming

In if structure the block of statements is executed only,

When the condition is true

Which of the following function(s) is/are included in stdlib.h header file?

1 and 2 only

Dealing with structures and functions passing by reference is the most economical method

True

Pointer is a variable which store,

Memory Address

Preprocessor program perform its function before phase takes place.

Compiling

Which of the following can not be a variable name?

10area

Which looping process is best, when the number of iterations is known?

all looping processes require that the iterations be known

Which character is inserted at the end of string to indicate the end of string?

null

How many bytes are occupied by declaring following array of characters?
`char str[] = "programming";`

12

Which of the following header file defines the `rand()` function?

stdlib.h

Commenting the code

Makes a program easy to understand for others.

What's wrong with this for loop?
`for (int k = 2, k <= 12, k++)`

the commas should be semicolons

For which array, the size of the array should be one more than the number of elements in an array?

char

To Which category of the software "Compiler and Interpreter" belongs?

What is the result of the expression $x = 2 + 3 * 4 - 4 / 2$

12

**first multiplies $3*4 = 12$ then Division $4/2 = 2$
 $2+12-2 = 12$**

Question No: 19

Write a declaration statement for an array of 10 elements of type float. Include an

initialization statement of the first four elements to 1.0, 2.0, 3.0 and 4.0.

Question No: 20

by Attia Kundi

Write down the output of the following code?

```
int array[7], sum = 0;
for(int i=0;i<7;i++)
{
    array[i] = i;
    sum+= array[i];
}

cout<< " Sum = " <<sum;
```

ANS

21

Loop will run times starts from zero and add values from 1 to 6 which is equal to 21

What will be the output of the following segment of C++ code?

```
int A[5] = {1, 2, 3, 4};
int i;
for (i=0; i<5; i++)
{
    A[i] = 2*A[i];
    cout << A[i] << " ";
}
}
```

ANS

2 4 6 8 0

Loops will run 5 times as its starting from zero. It will multiply the value of each item in array as last time is not initialized so it will multiply it with zero to give zero as output

Question No: 22

(Marks: 10

)

Write a C++ program that will determine if a departmental store customer has exceeded the credit limit on a charge account.

Program should input the following facts in five variables

1. Account number
2. Balance at the beginning of month (Beginning balance)
3. total of all items charged by customer this month (charges)
4. total of all credits (credits)
5. allowed credit limit

Calculate the new balance

New balance = Beginning balance + charges – credits

Determine if new balance exceeds the allowed credit limit. For those customers whose credit limit is exceeded. The program should display the message “Credit

by Attia Kundi

Limit exceeded.”

In C/C++ the string constant is enclosed

In single quotes

In flow chart, the symbol used for decision making is,

Diamond

The data type before a function name represents its,

Return Type

The operator used to take the address of a variable is,

(e) &

The header file which is used for input and output is

iostream.h

When we are using command line argument(s), the default argument in C/C++ is/are

_____.

argc and argv

In C++, Integer calculation occurs in _____ bytes.

2 bytes

In the declaration of two dimensional array,

First index represents row and second represents column

The address operator (&) can be used with,

Variable

__ translates high level language program into machine language code

Compiler

Which of the following data type(s) can operate on modulus operator „%“?

float, int

What will be the result of the expression $z = x \% y$, if $x = 19$ and $y = 4$?

3

Which character is inserted at the end of string to indicate the end of string?

null

What will be the value of i and j in the following code segment?

```
int i, j ;
int x[5] = {2, 3, 4, 8, 9} ;
int *ptr
=&x[2];
i =
(*ptr)++
;
j = *ptr++ ;
```

i = 5, j = 5

Syntax of union is identical to _____

Structure

Let suppose

```
Union intorDouble{
    Int ival;
```

```

    Double charvar;

};

main(){
    intorDouble VAZ;
    int size ;

    size = sizeof(VAZ);
}

```

What will be the value of variable "size", if int occupies 4 bytes and double occupies 8 bytes?

- a) 2
- b) 4
- c) 8
- d) 1

To Which category of the software "Compiler and Interpreter" belongs?

System software

8

Q20

What will be the output of following code segment?

(3)

```

int x[5] = {2, 4, 5, 7, 1} ;
int *ptr =&x[2];
cout << (*ptr)++
<<" "; cout <<
*ptr++;

```

Detect and correct compile time error(s) in the following code.

Hints: The following code pass a value to a function by Reference and print the value before and after pass. (5)

```

int    test (int &x)
{
*x = *x + 200;

```

```
return x;  
}
```

```
main()  
{  
int x = 100;
```

```
cout <<" x = " <<x;
```

by Attia Kundi

```

    test (*x); // //invalid type of argument uniary

    cout<<endl<< " x = " <<x;

    getch();
    return 0;
}

```

ANS

The following function pass a reference to the function but in the main body the call mechanism is wrong. The * sign is not used with it .

X = 100 after the function completion x = 300;

Q22

Write a C/C++ program which calculates the aggregate of a student.

The aggregate can be calculated by the formula: (10)

$$\text{Aggregate} = (\text{Matrix_marks} \times 2 + \text{Fsc_marks} \times 4) / 24$$

If the aggregate of a student is less than 150 then the program should display message

“You can not be admitted to VU” otherwise display the message

“Congratulation! You admitted in VU “

ANS

```
#include<iostream.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    int m,f;
```

```
    float a;
```

```
    m=254;
```

```
    f=340;
```

```
    a = ((m*2) +
    (f*4))/24; if(a < 150)
```

```
    {
```

```
        cout <<"You can not be admitted to VU";
```

```
    }
```

```
    else
```

```

{
    cout << "Congratulation! You admitted in VU ";
}

//Aggregate = (Matrix_marks*2 + Fsc_marks*4) / 24

getche();
}

```

Q1

What is function of **cout** ?

- (a) To send data to printer
- (b) To read data from keyboard
- (c) To display message

To display output on the screen

Q2

In Flow chart process is represented by

Rectangle

- (a) Arrow symbol
- (b) Oval
- (c) Circle

Q3

&& is----- operator.

- (a) An arithmetic

Logical

- (b) Relational
- (c) Unary

Q4

An over Flow condition occurs when we try to assign a value to a variable which is,

Less than its maximum size

- (a) Greater than its maximum size
- (b) With in its range
- (c) Equal to its size

Q5

For dereferencing an array element using pointer, we use the operator

&

- (a) *
- (b) /
- (d) +

Q6

In the declaration of two dimensional array,

First index represents row and second represents column

(a) First index represents column and second represents row

by Attia Kundi

- (b) Both indexes represent rows
- (c) Both indexes represent column

Q7

Which of the following data type(s) can operate on modulus operator „%“?

float, int

- (a) float, double
- (b) int
- (c) char

Q8

Which of the following is the correct way to declare a variable x of integer type?

- (a) x int ;
- (b) integer x ;

int x;

- (c) x integer

Q9

Which of the following can not be a variable name?

- (a) area
- (b) _area

(e) 10area

- (c) area2

Q10

Which of the function call is *call by value* for the following function prototype? float area (int);

- (a) area(&num) ;

(f) area(num) ;

- (b) area(int num) ;
- (c) area(*num)

; Q11

Recursive functions are used when there is a repetitive pattern.

(g) True

- (a) False

Q12

What will be the range of numbers generated by function rand () % 9?

- (a) 0 to 9
- (b) 1 to 9

0 to 8

- (c) 1 to

8 Q13

What will be the correct syntax to declare two-dimensional array of float data type?

- (a) float arr{2}{2} ;

(e) float arr[2][2] ;

- (b) float arr[2,2] ;

(c) float[2][2] arr
; Q14

When a function finishes its execution then,

(a) The control return to its Prototype

by Attia Kundi

(b) The control returns to its definition

(f) Control returns to statement following function call

(c) The compiler stop execution of whole program Q15

Consider the following statements to initialize a two-dimensional array.

i. `int arr[2][3] = {4, 8, 9, 2, 1, 6} ;`

iii. `int arr[][2] = {{4,8},{9, 2},{1, 6}} ;`

Which of the following option(s) are correct to initialize a two-dimensional array with 3 rows and 2 columns?

(a) (ii) only

(b) (iii) only

(ii) and (iii)

(c) (i)

and Q16

Editors are used to compile the code.

(a) True

False

Q17

What are global variables? (1)

ANS

Global variables are those that are defined outside of main. It can be accessed in entire program.

Q18

Is it possible to evaluate the size of structure, if yes then how? (1)

ANS

YES

```
#include <iostream.h>
```

```
#include <stdlib.h>
```

```
struct VehicleParts
```

```
{
```

```
int
```

```
wheels;
```

```
int seats;
```

```
VehicleParts()
```

```
{
```

```
cout << "\n VehicleParts - default constructor";
```

```
}
```

```
VehicleParts(int wheels, int seats)
```

```
{
    this->wheels = wheels; this->seats = seats; cout << "\n
    VehicleParts - parameterized constructor";
}
```

Q19

Write down the general syntax of switch statement. (22)

ANS

```
switch ( variable/expression )
```

```
{
```

```
case constant1 : statementLlist1 ; case constant2: statementLlist1 ; default: statementLlist1 ;
}
```

Q20

What will be the output of following code segment? (3)

```
int x[5] = {2, 4, 5, 7, 1} ;
```

```
int *ptr =&x[2];
```

```
cout << (*ptr)++
```

```
<<" "; cout <<
```

```
*ptr++;
```

ANS 5 6 Q22

What is the difference between compiler and interpreter? (5)

ANS

compiler and interpreter are system software but the difference is that the compiler compile the whole program but the interpreter is compiled line by line. Compiler are efficient in performance.

Q23

(10)

Write a recursive function that takes three arguments (an integer array, starting subscript 's' and ending subscript 'e').

In first recursive call, the function should display the array from subscript 's' (s = 0) to 'e' (e = size of array). In each successive call, the function should print the array from index s+1 to e. The function should stop processing and return when starting subscript becomes equal to ending subscript.

For example, if user enters values for array 2, 3, 4, 5, 6 then the recursive function must display the following output.

2 3 4 5 6

3 4 5 6

4 5 6

5 6

6

ANS

```
#include<iostream.
h>
#include<conio.h>

void recursive(int
[],int,int); void main()
{
    int array[5];
    for(int
    i=0;i<5;i++)
    {
        cout<<"\nEnter the "<<i<<" Index number
        .:"; cin>>array[i];
    }
    recursive(array,0,4); //0 is the starting index and 4 is the ending subscript getche();
}

void recursive(int arr[],int s,int e)
{
    if(s!=e+1)
    {
        for(int i=s;i<=e;i++)
            cout<<arr[i]<<"\t";

        cout<<endl;
        s++;

        recursive(arr,s,e);          //Recursive call
```

```

    }
}

```

Q1

What is function of **cout** ?

- (a) To send data to printer
- (b) To read data from keyboard
- (c) To display message

To display output on the screen

Q2

<, <=, >, >= are called ----- operators.

- (a) Logical
- (b) Arithmetic

Relational

- (c) Conational

Q3

In while loop the loop counter must be initialized,

- (a) With in the loop
- (e) Before entering the loop**
- (b) At the end of the loop
- (c) None of the given options

Q4

Data Size of the file is always _____ the actual size of the file.

- (f) Greater than**
- (a) Equal to
- (b) Less than or equal to
- (c) None of the above

Q5

The precedence of * is higher than dot operator (.)operator

- (g) True**
- (a) False

Q6

Let ptr1 and ptr2 are pointer variables that points to integer data type then which one of the following arithmetic is allowed,

- (a) ptr1 + ptr2
- (b) ptr1 - ptr2
- (h) ptr1 * ptr2**
- (c) ptr1 / ptr2

Q7

Word processor is

- (a) Operating system
- (i) Application software**
- (b) Device driver
- (c) Utility software

Q8

What will be the range of numbers generated by function `rand () % 9`?

(a) 0 to 9

(b) 1 to 9

(j) 0 to 8

(c) 1 to 8

Q9

How many bytes will the pointer ***intPtr*** of type ***int*** move in the following statement?

```
intPtr += 3;
```

(a) 3 bytes

(b) 6 bytes

12 bytes

(c) 24

bytes Q10

What will be the correct syntax to assign an array named ***arr*** of 5 elements to a pointer ***ptr***?

(a) `*ptr = arr ;`(b) `ptr = arr ;`**`*ptr = arr[5] ;`**(c) `ptr = arr[5]`

; Q11

What will be the output of the following code segment?

```
char *x = "programming" ;
```

```
cout << *(x+2) << *(x+3) << *(x+5) << *(x+8) ;
```

(a) prgm

(b) rorm

(e) ogai

(c) ramg

Q12

There is a pointer variable named ***ptr*** of type ***int*** then address of which type of variable the ***ptr*** will store in it?

(a) variable of type ***char***(b) variable of type ***short*****(f) variable of type *int***(c) variable of type ***double***

Q13

Flow charts explain the working of a program in pictorial format.

(g) True**(a) False**

Q14

The object _____ may be used both for file input and file output

(a) ***fstream***,**(h) *ifstream*,**

- (b) ofstream,
- (c) none of the given

options. Q15

Most efficient method of dealing with structure variables is to define the structure globally

(i) True

- (a) False

Q16

If a variable is passed by value to a function and the function makes some changes to that variable then it

(i) does not affect the original variable

- (a) affects the original variable
- (b) causes syntax error
- (c) None of the given options

Q17

How does elements of two-dimensional array store into memory? (1)

ANS

Two dimensional arrays are also stored in liner order. As like one dimensional array.

Q18

What is a structure? (1)

ANS

In structure, we introduce a new data type. "A structure is a collection of variables under a single name. These variables can different types, and each has a name that is used to select it from the structure"

Q19 (2)

When a pointer is incremented then how many bytes will it move to change its address?

ANS

If an integer occupies four bytes in the memory, then the `yptr++`; will increment its value by four.

Q20

What happens when we increment a pointer? (3)

ANS

become 11. The increment of a pointer depends on its data type. The data type, the pointer points to, determines the amount of increment. In this case, *yptr* is an integer pointer. Therefore, when we increment the *yptr*, it points to the next integer in the memory. If an integer occupies four bytes in the memory, then the *yptr++*; will increment its value by four. This can be understood from the following example.

Q
2
1

What are the advantages of random access file over sequential access file? (5)

Q22

Write a C/C++ program which calculates the aggregate of a student.

The aggregate can be calculated by the formula: (10)

$$\text{Aggregate} = (\text{Matrix_marks} \times 2 + \text{Fsc_marks} \times 4) / 24$$

If the aggregate of a student is less than 150 then the program should display message

“You can not be admitted to VU” otherwise display the message

“Congratulation! You admitted in VU “

ANS

```
#include<iostream.h>
```

```
#include<conio.h>
```

```
main()
```

```
{
```

```
    int m,f;
```

```
    float a;
```

```
    m=254;
```

```
    f=340;
```

```
    a = ((m*2) +  
    (f*4))/24; if(a < 150)
```

```
    {
```

```
        cout <<"You can not be admitted to VU";
```

```
    }
```

```
else
{
    cout <<"Congratulation! You admitted in VU ";
}

//Aggregate = (Matrix_marks*2 + Fsc_marks*4) / 24

getche();
}
```

Q1

In C/C++ language the header file which is used to perform useful task and manipulation of character data is

(a) cplex.h

ctype.h

(b) stdio.h

(c) delay.h

Q2

The header file which is used for input and output is

(a) maths.h

(b) string.h

iostream.h

(c) ctype.h

Q3

Suppose `int multi[5][10]`; when we are using `**multi`, it means,

(a) Single dereferencing

(b) Single referencing

Double referencing

(c) Double dereferencing

Q4

To access the data members of structure ____ is used.

dot operator (.)

(a) * operator

(b) € operator

(c) None of given.

Q5

The precedence of * is higher than dot operator (.) operator

(f) True

(a) False

Q6

Which of the following is the starting index of an array in C++?

(g) 0

(a) 1

(b) -1

(c) any number

Q7

When a call to function statement is encountered,

(h) The control transfers to its Prototype

(a) The control transfers to its definition

(b) Control returns to statement following function call

(c) The compiler stops execution of whole program

Q8

A function must always return value.

(a) True

(i) False

Q9

If an array has 100 elements, what is allowable range of subscripts? (a) 0 –

99

(b) 1 – 99

(c) 0 – 100

(d) 1

– 100

Q10

If a pointer appears on left hand side of an assignment operator then right side of that assignment operator must be,

(a) Variable name

Address of variable

- (b) Variable value
- (c) Constant

Q11

Editors are used to compile the code.

- (a) True

False

Q12

Which bitwise operator returns false if both bits are 1?

- (a) AND
- (b) XOR
- (c) NOT

OR

Q13

What does !(7) evaluate to in decimal where ! is a NOT operator?

7

- (a) 8
- (b) 9
- (c) 1

Q14

Structures cannot be passed as Function Parameters

- (a) True

False

Q15

When break statement is encountered in a loop body it,

Transfers the control outside from current loop

- (a) Transfers the control outside from current program
- (b) Enforces the next iteration of loop
- (c) Generates compile time

error. Q16

A union is a user-defined data type that contains only___from its list of members at a time.

- (a) One object

Two objects

- (b) Three objects
- (c) None of the given options

Q17

Why programming is important? Describe in ONE line. (1)

“A program is a precise sequence of steps to solve a particular problem.”

Q18

Which bitwise operator returns true if both bits are different and returns false if both bits are same? (1)

ANS

1. !=
2. ==

Q19

Why we close a file after use? (2)

**A
N
S**

To save our data stored on file. Also this process makes our program fast and reliable.

Q20

Can you use an assignment operator to assign the value of one C-string to another? (3)

ANS

Yes we can assign value c-string to another

ie char a={"I Love Pakistan"};

char b={"I Love Pakistan"};

Q21

The statement int Name [2][2]; define a 2x2 array, Write the code which read data from keyboard for this array. (5)

ANS

```

for(int i=0; i<2; i++)
{
    for (w=0; w<2; w++)
    {
        cin >>Name[i][w];
    }
    cout <<endl;
}
for(int i=0; i<2; i++)
{
    for (w=0; w<2; w++)
    {
        cout<<Name[i][w]<<" ";
    }
    cout <<endl;
}

```

Q22

Write a program which reads a text file "**PlayersInfo.txt**" residing in the current directory. Open the file **PlayersInfo.txt** in read mode and assign these values to the struct **Player**; assume order of the data in the file to be exactly the same as the

order of struct attributes. The struct **Player** has following attributes

- i) Name
- ii) Height
- iii) Age
- iv) Score
- v) Game

After reading the file and assigning values to the struct, in the end close the file

PlayersInfo.txt.

(10)

Q1

The size of int data type is

- (a) 1 bytes
- (b) 2 bytes**
- (c) 3 bytes
- (d) 4 bytes

Q2

When the logical operator AND (&&) combine two expressions exp1 and exp2 then the result will be true only,

When both exp1 and exp2 are true

- (a) When both exp1 and exp2 are false
- (b) When exp1 is true and exp2 is false
- (c) When exp1 is false and exp2 is true

Q3

The correct syntax of do-while loop is,

- (a) (condition) while; do { statements; };
- (b) { statements; } do-while ();
- (c) while(condition); do { statements; };
- (d) do { statements; } while (condition);**

Q4

_____provide communication channels between files and program.

- (a) Streams
- (b) Language like C++
- (c) Function seekg()**
- (d) None of the above

Q5

All elements of an array must be of,

- (a) different data type
- (b) float data only
- (c) character data only
- (d) same data type**

Q6

Function seekg() takes_____parameter(s).

- (a) 0

1

- (b) 2
 - (c) 3
- Q7

Structures help to define program-specific_____.

functions

- (a) datatypes
 - (b) Arithmetic operations
 - (c) None of the given options.
- Q8

In the declaration of two dimensional array,

First index represents row and second represents column

- (a) First index represents column and second represents row
 - (b) Both indexes represent rows
 - (c) Both indexes represent column
- Q9

What will be the result of arithmetic expression $6 + 27/3 * 3$?

33

- (a) 45
- (b) 9
- (c) 3

0 Q10

Which of the function call is **call by value** for the following function prototype?

```
float area (int);
```

- (a) `area(&num) ;`
- (f) **`area(num) ;`**
- (b) `area(int num) ;`
- (c) `area(*num)`

; Q11

How many bytes are occupied by declaring following array of characters?

```
char str*+ = "programming";
```

- (a) 10
- (b) 11
- (c) **12**
- (d) 1

3 Q12

What will be the correct syntax to assign an array named **arr** of 5 elements to a pointer **ptr**?

- (a) `*ptr = arr ;`
- (b) `ptr = arr ;`
- (e) **`*ptr = arr[5] ;`**
- (c) `ptr = arr[5] ;`

Q13

Let ptr1 and ptr2 are pointer variables then which of the following

arithmetic operation is allowed on pointers?

- (a) Addition
- (b) Subtraction
- (c) Multiplication

All of the above

Q14

The variables having a name, type and size are just like empty boxes.

True

- (a) False

Q15

When break statement is encountered in a loop body it,

- (a) **Transfers the control outside from current loop**
- (b) Transfers the control outside from current program
- (c) Enforces the next iteration of loop
- (d) Generates compile time

error. Q16

If two programs **a** and **b** are trying to open a file **xyz.txt** at approximately same time then

(e) Both programs will generate error

- (a) One of them will succeed in opening that file and other will fail
- (b) Both programs will open the file
- (c) One of the program will re-

start Q17

What is meant by array manipulation? (1)

Q18

What will be the value of x after the execution of the following code segment? (1)

```
int x=10;
int y =30;

int *xptr = &x; x
= *xptr + 10;
```

Q19

(2)

What is the output of the code given below? void main()

```
{
int a=10,b=20;
char x=1,y=0;
if(a,b,x,y)
{
cout << "EXAM";
```

```
}
}
```

Q
2
0

What is Overflow condition? (3)

ANS

When we try to store larger information in a variable, than a data type can store, overflow condition occurs.

Q21

Write code which read a string not greater than 20 characters from keyboard stored it in an array Name and display it on the screen. (5)

ANS

```
char string[19];
cout<<"Enter a
string"; cin >>
string;
```

Q22 (10)

Write a C++ program which contains a user-define function named **convertHeight** which takes height of person in centimeter as an argument. This function converts the height in centimeter into feet and inches and displays them on the screen.

Program should prompt the user to enter height in centimeter and pass it to function **convertHeight** as an argument which displays height in feet and inches.

Hint:

1 foot = 12 inches
1 inch = 2.5 cm

ANS

```
#include<iostream.h>
#include<conio.h>
>

void convertHeight
(int); main()
{
int hight;
cout <<"Enter the High of the Person = "; cin
>> hight;
```

```
convertHeight(hight);
```

```
getche();
}
```

```
void convertHeight (int f)
{
//1 foot = 12 inches
//1 inch = 2.5 cm

int i,c;
i = f*12;
c = i*2.5;
cout <<"high in Feet = "<<f<<endl; cout
<<"high in inches = "<<i<<endl; cout
<<"high in CM = "<<c<<endl;

}
```

Q1

The data type of size one byte is

- (a) char
- (b) int
- (c) long
- (d) double

Q2

If Num is an integer variable then Num*= 4; means,

- (a) Multiply Num 4 times
- (b) Multiply 4 with Num and display
- (f) Multiply 4 with Num and assign the result to Num**
- (c) Add 4 with Num

Q3

Member function tellg() returns the current location of the _____ pointer.

- (a) tellptr()**
- (b) write()
- (c) seekg()
- (d) get()

Q4

If we want to store a string "abc" in an array str then the size of this array must be at least,

- (e) 2**
- (a) 3
- (b) 4
- (c) 5

Q5

Pointer is a variable which store,

- (a) Values
- (b) Data
- (f) Memory Address**
- (c) Data Type

Q6

C is widely known as development language of _____ operating system.

- (a) Linux
- (g) Windows**
- (b) Unix
- (c) Mac

OS Q7

C++ is a case-sensitive language

- (h) True**
- (a) False

Q8 What is the output of the following code?

```
for (int a = 1; a <= 1; a++) cout << a++; cout << a;
```

- (a) 22
- (b) 12**
- (c) 23
- (d) 13

Q9

A continue statement causes execution to skip to (a) the return 0; statement

- (b) the first statement after the loop
- (c) the statements following the continue statement**
- (d) the next iteration of the loop**

Q10

If there is more than one statement in the block of a for loop, which of the following must be placed at the beginning and the ending of the loop block?

- (a) parentheses ()
- (b) braces { }**
- (c) brackets []
- (d) arrows <

> Q11

What will be the correct syntax for the following function call?

```
float square (int&);
```

- (a) square (int num);**
- (e) square (&num);**
- (b) square (num);**
- (c) square**

```
(*num);
```

Q12

Which of the following is the correct way to assign an integer value 5 to element of a matrix say „m“ at second row and third column?

(f) **(a) m[2][3] = 5 ;**

(b) m[3][2] = 5 ;

(c) m[1][2] = 5 ;

(d) m[2][3] = „5“; Q13

Class can be defined as

A class includes both data members as well as functions to manipulate that data

_____operator uses the sign ^

Exclusive OR

For console input and output we use_____.

conio.h header file

Header file provide_____so the program running on one operating system can run without an error on the other system

Portability

Constructor has_____.

The same name as of class

The object code of our program is combined with the_____.

Object code of the library functions

The malloc functions takes_____argument(s)

Two

The function overloading requires_____.

The argument list to be different

Bugs can occur due to_____.

Uninitialized data

The name of the destructor is the same as that of a class proceeding with a_____.

~sign

The main advantage of function overloading is that_____.

The program becomes more reliable

How many dimensions does n-dimensional array has?

(g) n dimensions

(a) $2n$ dimensions

(b) $(n+1)$ dimensions

(c) $(n-1)$

dimensions Q14

Consider the following code segment. What will be the output of following code?

```
int addValue
(int *a){ int b =
(*a) + 2; return
b ;
ma
in
()
{
int
x
=6
;
cout << x << " , " ;
cout << addValue(&x)
<< " , " ; cout << x ;
}
```

(a) 6,8,6

(b) 6,6,8

(c) 6,8,8

(d)

6,6,6 Q15

If most significant bit of un-signed number is 1 then it represents a positive number.

(a) **True**

(b) Fals

e Q16

When we declare a multidimensional array the compiler store the elements of multidimensional array in the form of,

by Attia Kundi

(a) Contiguous memory location

(c) **Matrix**

(b) Columns

(c) Row

s Q17

What is the functionality of the function: `char *strncat (char *s1, const char *s2, size_t n)` (1)

ANS

_____ of `s1`. The value of `s1` is returned.

Q1

8

Write a piece of code that outputs three values of type int, long and float to a stream. (1)

ANS

```
main()
{
    int a;
    long b;
    float c;
    a = 10;
    b = 124568979;
    c = 6.57;
    cout << a << "\t" << b << "\t" << c;

    getch();
}
```

Q19

Which bit of the number is used as a sign bit? (2)

ANS

The most significant bit is used as a sign bit.

Q20

What is difference between single-dimensional and multi-dimensional array?

ANS

Single dimensional array used to stored lists and multi dimentional arrays used to stored value as tabler formate i.e matrix.

Q2

1

Write down the C++ program that calculates the Zakat on the amount entered by the user

Note: Zakat is 2.5% of the given amount (5)

ANS

```
main()
{
    int amount;
    float zakat;
    cout <<"Enter the Amount = ";
    cin >>amount;
    zakat = (amount*2.5)/100;
    cout <<"Amount = "<<amount<<endl; cout
    <<"Amount = "<<zakat<<endl; getch();
}
```

Q22

What is meant by scope of identifiers? Differentiate between different scope of identifiers and explain them with examples? (10)

Scope of Identifiers

An 'Identifier' means any name that the user creates in his/her program. These names can be of variables, functions and labels. Here the scope of an identifier means its visibility. We will focus Scope of Variables in our discussion.

Now this variable „i“ can be used in any statement inside the function func1(). But consider this variable being used in a different function like:
void func2()

```
{

int k = i + 4; //Compilation error
}
```

The variable „i“ belongs to func1() and is not visible outside that. In other words, „i“ is local to func1().

Q1

What is function of **cout** ?

- (a) To send data to printer
- (b) To read data from keyboard
- (c) To display message
- (d) To display output on the screen**

Q2

For one byte there are ____ combinations of values that can be stored in computer.

- (A) 2^6
- (B) 2^7
- (C) 2^8
- (D) 2^4

Q3

_____ provide communication channels between files and program.

- (a) Streams
- (b) Language like C++
- (c) Function seekg()**
- (d) None of the above

Q4

The data type before a function name represents its,

- (e) Return Type**
- (a) Function data
- (b) Function arguments
- (c) Function name

Q5

In C/C++ language when an array is passed to a function then by default its passing mechanism is,

- (a) Call by value
- (f) Call by Reference**
- (b) It depends on type of array
- (c) It depends on the return type of function.

Q6

Array is a data structure which store

- (a) Memory addresses
- (g) Variables**
- (b) Data Type
- (c) Data

Q7

If there is more than one statement in the block of a for loop, which of the

following must be placed at the beginning and the ending of the loop block?

- (a) parentheses ()
- (b) braces { }**
- (c) brackets []
- (d) arrows < >

Q8

Array is passed by value to a function by default.

- (a) True
- False**

Q9

Which of the following is the correct function call having array named *student* of 10 elements as a parameter.

- (a) addRecord(student[]) ;
- (b) addRecord(student) ;
- addRecord(student[10]) ;**
- (c) addRecord(*student) ;

Q10

What will be the correct syntax for initialization of pointer *ptr* of type int with variable *x*?

- (a) int ptr = &x ;
- (b) int ptr = x ;
- (g) int *ptr = &x ;**
- (c) int ptr* = &x ;

Q1

1 What will be the correct syntax for initialization of pointer *ptr* with string "programming"?

- (a) char ptr = 'programming' ;
- (b) char *ptr = "programming" ;**
- (c) char *ptr = 'programming' ;
- (d) *ptr = "programming"

; Q12

The condition in while loop may contain logical expression but not relational expression.

- (e) True**
- (a) False

Q13

We want to access array in random order which approach is better?

- (f) Pointers**
- (a) Array index
- (b) Both pointers and array index are better
- (c) None of the given

options. Q14

Single line comments explaining code would be preceded like in the following example.

- (a) /*
- (g) //**
- (b) /

Q15

Function write() takes _____ as parameter(s).

- (a) String of pointer type
- (b) String and no. of bytes to be written
- (c) Pointer array of characters and delimiter

String of variable lengths, no. of bytes to be read and flags

Q16

Structure is a collection of _____ under a single name.

- (a) **Only Functions**
- (b) Only Variables
- (c) Both Functions and Variables
- (d) None of the given options

Q17

What will be the correct syntax to initialize a pointer „ptr“ with two-dimensional array „m“? (1)

ANS

```
int m[2][2];
int * ptr;
ptr = *m;
```

Q18

Which one of the loop (while or do-while) must be used if it is necessary to execute a loop at least once? (1)

ANS

do-while loop

Q19

Identify each of the following function as string conversion function or string manipulation function. (2)

```
double atof(const char *nptr)
char *strcpy ( char *s1, const char
*s2) int atoi(const char *nptr)
```

ANS

1. **double atof(const char *nptr)** ---€ Converts the string nPtr to double.
2. **char *strcpy (char *s1, const char *s2)** -€ Copies string s2 into character array s1. The value of is returned.
3. **int atoi(const char *nptr)**--€ Converts the string nPtr to int.

Q20

What is difference between single-dimensional and multi-dimensional array? (3)

ANS

Single dimensional array used to store lists and multi-dimensional arrays used to store values as table format i.e. matrix.

Q21

What will be the output of the following code segment?(5)

```
int num[10] = {2, 3, 5, 8, 9, 10, 12, 15, 19, 20};
int *ptr = num;
for (int i=0; i<10; i+=2){
    cout << *(ptr+i) << " ";
}
```

ANS

2, 5, 9, 12,

19, Q22

Write a C++ program which contains a user-defined function named **convertHeight** which takes height of person in centimeter as an argument. This function converts the height in centimeter into feet and inches and displays them on the screen.(10)

Program should prompt the user to enter height in centimeter and pass it to function **convertHeight** as an argument which displays height in feet and inches.

Hint:

1 foot = 12 inches
1 inch = 2.5 cm

ANS

```
#include<iostream>
using namespace std;
#include<conio.h>

void convertHeight
(int);

int main()
{
    int height;
    cout << "Enter the Height of the Person = "; cin
    >> height;
    convertHeight(height);

    getch();
}

void convertHeight (int f)
{
    //1 foot = 12 inches
```

```
//1 inch = 2.5 cm
```

```
int i,c;
i = f*12;
c = i*2.5;

cout <<"high in Feet = "<<f<<endl; cout
<<"high in inches = "<<i<<endl; cout
<<"high in CM = "<<c<<endl;

}
```

Q1

Compiler is a

- (a) **System software**
- (b) Application Software
- (c) Driver
- (d) Editor

Q2

In while loop the loop counter must be initialized,

- (a) With in the loop
- (e) **Before entering the loop**
- (b) At the end of the loop
- (c) None of the given options

Q3

If Num is an integer variable then Num*= 4; means,

- (a) Multiply Num 4 times
- (b) Multiply 4 with Num and display
- (f) **Multiply 4 with Num and assign the result to Num**
- (c) Add 4 with Num

Q4

In C/C++ ,the arguments are passed by_____to a function by default .

- (a) reference
- (g) **value**
- (b) data
- (c) type

Q5

Disks is divided into _____with power of_____.

- (a) **Chunks, 2n**
- (b) Blocks, n^2
- (c) Blocks, 2^n
- (d) Chunks, n^2

Q6

C is widely known as development language of_____operating system.

- (a) Linux
- (h) **Windows**
- (b) Unix

(c) Mac OS

Q7

Assignment operator „=“ is a

(a) Unary operator

(b) Binary operator

(c) Ternary operator

(d) None of the given options

Q8

Consider the following code segment. What will be the output of the following program?

```
int func(int);
int num = 10;
```

```
int
main(){
int num ;
num = 5 ;

cout << num ;

cout << func(num) ;
}

int func(int
x){ return
num ;
}
```

(a) 5, 5

(b) 10, 5

(c) 5, 10

(d) 10, 10

Q9

Name of an array is a constant pointer.

(a) True

(b) Fals

e Q10

What will be the correct syntax to assign an array named **arr** of 5 elements to a pointer **ptr**?

(a) *ptr = arr ;

(b) ptr = arr ;

(c) *ptr = arr[5] ;

(d) ptr = arr[5]

; Q11

If there are $2^{(n+1)}$ elements in an array then what would be the number of

iterations required to search a number using binary search algorithm?

by Attia Kundi

- (a) n elements
- (b) (n+1) elements
- (c) 2(n+1) elements
- (d) $2^{(n+1)}$

elements Q12

In C/C++, null character is represented as

- (e) **\n**
- (a) \0
- (b) \t
- (c) \

r Q13

How many nested loop would be required to manipulate n-dimensional array?

- (a) 2n
- (b) **n**
- (c) n + 1
- (d) n -

1 Q14

What will be the correct syntax to access the value of fourth element of an array using pointer ptr?

- ptr[3]**
- (a) (ptr+3)
- (b) *(ptr+3)
- (c) Both 1 and

3 Q15

Single line comments explaining code would be preceded like in the following example.

- (a) /*
- //**
- (b) /
- (c) //

Q16

If a variable is passed by value to a function and the function makes some changes to that variable then it

does not affect the original variable

- (a) affects the original variable
- (b) causes syntax error
- (c) None of the given options

Q17

What is meant by *num and &num? [1]

ANS * num is a pointer and &num is a reference to that pointer. Q18

Suppose there is a pointer to structure ***sPtr**. How can we access the data member „name“ with sPtr? [1]

ANS

Structure data members using pointers Using the * operator;
(*sptr).name

Q19

ANS Why we close a file after use? [2]

To save our data stored on file. Also this process makes our program fast and reliable

Q20

Define Flow
chart. [3] ANS

Flow Chart

A flow chart is a pictorial representation of a program. There are labeled geometrical symbols, together with the arrows connecting one symbol with other. A flow chart helps in correctly designing the program by visually showing the sequence of instructions to be executed.

Q21

Write down the function definition if we want to pass the arguments to a function by reference without changing the values stored at that addresses. [5]

Q22

From writing to execution of the program following software are used explain for what purpose each is used. [10]

Editor
Compiler/Interpreter
Linker
Loader

ANS

Editors is a tool for writing the code of a program. For this purpose we used Editors in which we write our code. We can use word processor too for this, but word processors have many other features like bold the text, italic, coloring the text etc, so when we save a file written in a word processor, lot of other information including the text is saved on the disk. For programming purposes we don't need these things we only need simple text. Text editors are such editors which save only the text which we type. So for programming we will be using a text editor

Compiler and Interpreter

Compilers translate the English like language (Code written in C) into a language (Machine language) which computers can understand. The Compiler reads the whole program and translates it into machine language completely. The difference between interpreter and compiler is that compiler will stop translating if it finds an error and there will be no executable code generated whereas Interpreter will execute all the lines before

error and will stop at the line which contains the error. So Compiler needs syntactically correct program to produce an executable code. We will be using compiler in our course As we write the code in English and we know that computers can understand only 0s and 1s. So we need a translator which translates the code of our program

into machine language. There are two kinds of translators which are known as Interpreter and Compilers. These translators translate our program which is written in C-Language into Machine language. Interpreters translates the program line by line meaning it reads one line of program and translates it, then it reads second line, translate it and so on. The benefit of it is that we get the errors as we go along and it is very easy to correct the errors. The drawback of the interpreter is that the program executes slowly as the interpreter translates the program line by line. Another drawback is that as interpreters are reading the program line by line so they cannot get the overall picture of the program hence cannot optimize the program making it efficient.

Linker Most of the time our program is using different routines and functions that are located in different files, hence it needs the executable code of those routines/functions. Linker is a tool which performs this job, it checks our program and includes all those routines or functions which we are using in our program to make a standalone executable code and this process is called Linking

Loader after a executable program is linked and saved on the disk and it is ready for execution. We need another process which loads the program into memory and then instruct the processor to start the execution of the program from the first instruction (the starting point of every C program is from the main function). This processor is known as loader. Linker and loaders are the part of development environment. These are part of system software.

Q1

There are mainly -----types of software

- (a) **Two**
- (b) Three
- (c) Four
- (d) Five

Q2

<, <=, >, >= are called ----- operators.

- (a) Logical
- (b) Arithmetic
- (e) **Relational**
- (c) Conational

Q3

In order to get 256 from the number 2568 we divide this number by 10 and take,

- (a) Its remainder
- (b) The number

Its quotient

(c) Its divisor

Q4

If `int x = 10;` then the value of `x/= 3;` will be,

(a) 10

3

(b) 13

(c) 1

Q5

How many parameter(s) function `getline()` takes?

(a) 0

(b) 1

(c) 2

3

Q6

Suppose `int multi[5][10];` when we are using `**multi`, it means,

(a) Single dereferencing

(b) Single referencing

(c) Double referencing

(d) Double dereferencing

Q7

To access the data members of structure ____ is used.

dot operator (.)

(a) * operator

(b) € operator

(c) None of given.

Q8

There is NO difference between **bitwise AND operator (&)** and **Logical AND (&&) operator.**

True

(a) False

Q9

Which of the following data type(s) can operate on modulus operator `„%“`?

float, int

(a) float, double

(b) int

(c) cha

r Q10

What's wrong with this while loop?

```
while( (i < 10) && (i > 24))
```

(a) the logical operator `&&` cannot be used in a test condition

(b) the while loop is an exit-condition loop

the while loop is an exit-condition loop

(c) the test condition is always true

Q1

1 The switch structure is a _____ construct

- (a) single-selection
- (b) bi-selection

multiple-selection

- (c) unconditional

Q12

Keyword „array“ must be used to declare an array.

(a) True

(b) Fals

e Q13

What will be the correct syntax to declare two-dimensional array of float data type?

- (a) float arr{2}{2} ;

float arr[2][2] ;

- (b) float arr[2,2] ;

- (c) float[2][2] arr

; Q14

When an array element is passed to a function then this array element is passed to the function.

By reference

- (a) By data type

- (b) By value

- (c) By

data Q15

Which of the following operator is used to access the value of variable pointed to by a pointer?

- (a) * operator

- (b) -> operator

- (c) && operator

& operator

Q16

Paying attention to detail in designing a program is _____

- (a) Time consuming

- (b) Redundant

Necessary

- (c) Somewhat Good

Q1

7

How does elements of two-dimensional array store into memory? (1)

Two dimensional arrays are also stored in liner order. As like one dimensional array.

Q18

Which strategy is used by binary search algorithm to search a number? (1)

ANS

„divide and conquer“ strategy is applied.

Q19

Write down the general syntax of switch statement. (2)

ANS

switch (*variable/expression*)

{

```
case constant1 : statementLlist1 ; case constant2: statementLlist1 ; default: statementLlist1 ;
}
```

Q20

What is a Linker? (3)

Linker Most of the time our program is using different routines and functions that are located in different files, hence it needs the executable code of those routines/functions. Linker is a tool which performs this job, it checks our program and includes all those routines or functions which we are using in our program to make a standalone executable code and this process is called Linking

Q21

What are similarities and differences between Structures and Unions? (5)

Structure

In structures, the data members are public by default. It means that these are visible to all and anyone can change them. Is there any disadvantage of this? Think about the date.

syntax

struct student

{

char

name[60];

char

address[100];

float GPA;

};

Unions We have another construct named union. The concept of union in C/C++ is: if we have something in the memory, is there only one way to access that memory location or there are other ways to access it. We have been using int and char interchangeably in our programs. We have already developed a program that prints the ASCII codes. In this program, we have stored a char inside an integer. Is it possible to have a memory location and use it as int or char interchangeably? For such purposes, the construct union is used. The syntax of union is:

union intOrChar

{

i

n

t

```

i
;
c
h
a
r
c
;
};

```

Q22

Differentiate between C and c++. (10)

1. C was the C++ predecessor. As its name implies, a lot of C remains in C++. Although not actually being more powerful than C.
2. C++ allows the programmer to more easily manage and operate with Objects, using an OOP (Object Oriented Programming) concept
3. C++ allows the programmer to create classes, which are somewhat similar to C structures. However, to a class can be assigned methods, functions associated to it, of various prototypes, which can access and operate within the class, somewhat like C functions often operate on a supplied handler pointer.
4. Although it is possible to implement anything which C++ could implement in C, C++ aids to standardize a way in which objects are created and managed, whereas the C programmer who implements the same system has a lot of liberty on how to actually implement the internals, and style among programmers will vary a lot on the design choices made
5. In C, some will prefer the handler-type, where a main function initializes a handler, and that handler can be supplied to other functions of the library as an object to operate on/through. Others will even want to have that handler link all the related function pointers within it which then must be called using a convention closer to C++.
6. C++ applications are generally slower at runtime, and are much slower to compile than C programs. The low-level infrastructure for C++ binary execution is also larger. For these reasons C is always commonly used even if C++ has a lot of popularity, and will probably continue to be used in projects where size and speed are primary concerns, and portable code still required (assembly would be unsuitable then).

Q1

The remainder (%) operator is

- (a) A logical operator
- An arithmetic operator**
- (b) A relational operator
- (c) A division operator

Q2

If `int sum = 10;` then the value of the statement `sum = sum + 3 ;` is ,

- (a) 7
- (b) Illegal statement
- (c) Garbage value

13

Q3

Which of the following function(s) is/are included in **ctype.h** header file?

- (a) `isdigit(int c)`
- (b) `isxdigit(int c)`
- (c) `tolower(int c)`

All of the above

Q4

In C/C++ which of the following header file is used for string manipulation?

stdlib.h

- (a) `string.h`
- (b) `strings.h`
- (c) `styp.h`

Q5

_____ provide communication channels between files and program.

- (a) Streams
- (b) Language like C++

Function seekg()

- (c) None of the above

Q6

___ translates high level language program into machine language code

- (a) Debugger
- (b) Editor

Compiler

- (c) Linker

Q7

Which of the following data type(s) can operate on modulus operator „%“?

float, int

- (a) float, double
- (b) int
- (c) char

Q8

C++ is a case-sensitive language

True

- (a) False

Q9

To include code from the library in the program, such as `iostream`, a directive would be called up using this command.

- (a) `#include "iostream.h"`
- (b) `include <iostream.h>`
- (c) `include <iostream.h>`

#include <iostream.h>

Q10

What will be the range of numbers generated by function `rand () % 9`?

(a) 0 to 9

(b) 1 to 9

0 to 8

(c) 1 to 8

Q1

1 An array stores the numbers into consecutive memory locations.

True

(a) Fals

e Q12

Which of the following is the correct statement for the following declaration? `const int *ptr.`

(a) `ptr` is a constant

pointer **`ptr` is constant**

integer pointer

(b) `ptr` is a constant pointer to `int`

(c) `ptr` is a constant pointer to

int Q13

Which of the following header file defines the **`rand()`** function?

(a) `iostream.h`

(b) `conio.h`

`stdlib.h`

(c) `stdio.`

h Q14

Consider the following code segment. What will be the output of following code?

```
int addValue
(int *a){ int b =
(*a) + 2; return
b ;
}
main
()
{
int
x
=6
;
cout << x << " , " ;
cout << addValue(&x)
<< " , " ; cout << x ;
}
```

(a) 6,8,6

- (b) 6,6,8
- (c) 6,8,8
- (d) 6,6,

Q15

Identifier is a name that can be given to variables, labels and functions.

True

(a) Fal

s Q16

For which array, the size of the array should be one more than the number of elements in an array?

- (a) Int
- (b) Char
- (c) Double

float

Q17

Give a precise definition of **function** . (1)

AN

S **Functions** In C/C++, functions are a way of modularizing the code. A bigger problem is broken down into smaller and more manageable parts. There is no rule of thumb for the length of each part but normally one function's length is not more than one screen.

Q18

What will be the size of array if we initialize an array with declaration: **int arr[] = {0, 0, 0, 0};**? (1)

ANS

```
int
arr
[3]
;
```

Q1
9

What is the difference between **switch** statement and **if** statement. (2)

ANS

In switch statement only one variable can be tested on various condition but using if we can tested multi variables in single statement

Q20

Evaluate the following arithmetic expressions. (3)

- a) $X = 2 + 6 * 4 - 4 * 20 / 5 + 3 * 2$
- b) $Y = (6 * 7) - (2 + 3) * (3 - 1) + 5 * (3 + 1)$

ANS

- a) $X = 2 + (6 * 4) - ((4 * 20) / 5) + (3 * 2)$
 b) $Y = ((6 * 7) - (2 + 3) * (3 - 1) + (5 * (3 + 1)))$
 a) 16
 b) 52

Q21

What is the difference between = in C as compared to = used in algebra. (5)

ANS

In C = sign is used to assigned the value in algebra = sign is show that the both side are equal.

In C there must be a variable on the = sign and the right side of the = sign must be a arithmetic expression, variable or a value.

In algebra both side of the equation may or may not be expression. Q22

Write a program which consists of three variables Area, Per, Base, this program should find the area of triangle using the formula,
 (10)

$$\text{Area} = (\text{Base} * \text{Per})/2$$

```
#include<iostream.h>
#include<conio.h>
main()
{
    float Area, Per, Base;
    // Area = (Base * Per)/2
    Per = 5.0;
    Base = 2.5;
    Area = (Base * Per)/2; cout
    <<"Area = "<<Area;
    getch();
}
```

Q1

In C/C++ language the header file which is used to perform useful task and manipulation of character data is

- (d) cplex.h
ctype.h
 (e) stdio.h
 (f) delay.h

Q2

The header file which is used for input and output is

- (e) `maths.h`
- (f) `string.h`

by Attia Kundi

iostream.h

(g) ctype.h

Q3

Suppose `int multi[5][10]`; when we are using `**multi`, it means,

(e) Single dereferencing

(f) Single referencing

Double referencing

(g) Double dereferencing

Q4

To access the data members of structure____is used.

dot operator (.)

(e) * operator

(f) € operator

(g) None of given.

Q5

The precedence of * is higher than dot operator (.)operator

True

(c) False

Q6

Which of the following is the starting index of an array in C++?

0

(d) 1

(e) -1

(f) any number

Q7

When a call to function statement is encountered,

The control transfers**to its Prototype**

(e) The control transfers to its definition

(f) Control returns to statement following function call

(g) The compiler stops execution of whole program

Q8

A function must always return value.

(c) True

False

Q9

If an Array has 100 elements, what is allowable range of subscripts? (e) 0

- 99

1 - 99

0 - 100

1 - 100

Q10

If a pointer appears on left hand side of an assignment operator then right side of that assignment operator must be,

(d) Variable name

Address of variable

- (e) Variable value
- (f) Constan

t Q11

Editors are used to compile the code.

- (c) True

False

Q12

Which bitwise operator returns false if both bits are 1?

- (d) AND
- (e) XOR
- (f) NOT

OR

Q13

What does !(7) evaluate to in decimal where ! is a NOT operator?

7

- (e) 8
- (f) 9
- (g) 1

0 Q14

Structures cannot be passed as Function Parameters

- (c) True

False

Q15

When break statement is encountered in a loop body it,

Transfers the control outside from current loop

- (d) Transfers the control outside from current program
- (e) Enforces the next iteration of loop
- (f) Generates compile time

error. Q16

A union is a user-defined data type that contains only___from its list of members at a time.

- (e) One object

Two objects

- (f) Three objects
- (g) None of the given options

Q17

Why programming is important? Describe in ONE line. (1)

Q18

Which bitwise operator returns true if both bits are different and returns false if both bits are same? (1)

Q19

Why we close a file after use? (2)

Q20

Can you use an assignment operator to assign the value of one C-string to another? (3)

Q21

The statement `int Name [2][2];` define a 2x2 array, Write the code which read data from keyboard for this array. (5)

Q22

Write a program which reads a text file "**PlayersInfo.txt**" residing in the current directory. Open the file **PlayersInfo.txt** in read mode and assign these values to the struct **Player**; assume order of the data in the file to be exactly the same as the order of struct attributes. The struct **Player** has following attributes

- i) Name
- ii) Height
- iii) Age
- iv) Score
- v) Game

After reading the file and assigning values to the struct, in the end close the file **PlayersInfo.txt**. (10)

Q23

Q24

Write a program to convert upper case letter to lower case letter. (5)

Write a program to convert lower case letter to upper case letter. (5)

Q2

Write a program to convert upper case letter to lower case letter.

(5) Ans:

```
#include <iostream>
```

```
using std::cout;
```

```
using std::cin;
```

```
using std::endl;
```

```
#include <stdlib.h>
```

```
int main()
```

```
{
```

```
    char inputString[100];
```

```
    char lowerCase[100];
```

```
    cout<<"Please enter a string (maximum 100 characters): ";
```

```
    gets(inputString);
```

```
    int i=0;
```

```
    for(i=0; i<strlen(inputString); i++)
```

```
    {
```

```
        lowerCase[i] = tolower(inputString[i]);
```

```
    }
```

```
    lowerCase[i]='\0';
```

```
    puts(lowerCase);
```

```
    system("PAUSE");
```

```
    return 0;
```

```
}
```

Write a program to convert lower case letter to upper case letter. (5)

Ans:

```
#include <iostream>
```

```
using
```

```
std::cout;
```

```
using
```

```
std::cin;
```

```
using
```

```
std::endl;
```

```
#include
```

```
<stdlib.h>
```

```
int main()
```

```
{
```

```
    char
```

```
    inputString[
```

```
    100]; char
```

```
    upperCase[1
```

```
    00];
```

```
    cout<<"Please enter a string (maximum 100
```

```
    characters): "; gets(inputString);
```

```
    int i=0;
```

```
    for(i=0; i<strlen(inputString); i++)
```

```
    {
```

```
        upperCase[i] = toupper(inputString[i]);
```

```
    }
```

```
    upperCase[i]
```

=\0';

by Attia Kundi

```

puts(upperC
ase);
system("PA
USE");
return 0;
}

```

CS201

GRAND QUIZ FILE (NEW)

ALMOST 800 + MCQS

Which of the following is are serve word for data types in c/c++?

Int

C was developed with the help of ----- language

BCPL and B

--- is used for input file or for reading from the file.

Tellg()

Following is the declaration a -----array. `Int arr[2][2][2]`

3-demensional

! ($x < 3$) means in c++ that

X is greater than 3

Which of the following option is correct to compare the declaration of the function?

by Attia Kundi

```

----- (parameter1,parameter2)
{
// code to be executed
}
Void functionNAME

```

Ofstream is used for -----
Output file stream

By default, the starting index of an array in c++ is _____
0

How we declare an array of integer data whose size is 20 with array name "count".
Int count [20];

An object of ----- class can be created for saving the current position of any opened file before reading from the file.
Streampos

C++ is a ----- language.
High level

character is used to indicate the end of the string.
Null

In while loop, the loop counter is initialized at -----

Before entering the loop

What is the output of the following code if the 3rd case is true

```
Switch (var)
```

```
{Case „a“;
```

```
Cout<<"apple"<<endl;
```

Mango

Which of the following is used to perform bit-wise exclusive OR operation?

|

If $x = 11$, and $x \% = 3$, then value of x will be-----

2

A variable of character data type occupies -----bytes in memory.

1

Null character is represented by -----in c++?

\0

In computer system there are mainly-----type of softwares.

2

There is pointer variable named ptr of type float. Which type of variable address can be stored by ptr?

Float

A character is stored in the memory in -----

Byte

X is an integer variable; what does x++ means,

Add 1 in x value

When we call a function, its control-----

Transfers to its definition

'\0' takes ----- space in the memory.

1 byte

= operator is used for ----

Assigning values to variables

What is the use of ios::trunc mode?

To truncate an existing file to zero

X is an integer variable; what does x ++ means,

Add 1 in x value

_____operator uses the sign &

Bitwise AND operator

_____operator uses the sign |

Bit-wise OR operator

_____operator uses the sign ~

Not operator

_____operator uses the sign <<

Left shift operator

_____operator uses the sign >>

Right shift operator

What can be the resultant number when you AND two numbers 12 & bit-wise using & operator?

8

Structures use-----allocation.

Heap

What will be the size of the following character array?

Char name[] = "adeel";

6

A is an array of characters that can store number of

character specified.

String

---- statement is used to terminate the processing of a

particular case and exit from switch structure.

Break

----- will be used for enclosing function statements into a block?

{ }

How can we declare an array of characters whose size is

12 with array name "country"?

Char country[12];

Following is the declaration of a 2d array, how many columns are declared for this array.

Int arr[3][2];

2

If we assign 2.06721 to an integer variable x, what will be the output if we print x using cout << statement?

2

We can access a global variable ----

From anywhere in the program

We use ----- to include all those functions which are

located in different files.

Linker

A character is stored in the memory in ----

Byte

Which is not true regarding structure?

We can declare array of structure

Mr. ahmad sleeps thirty hours every day. This sentence is __

Grammatically correct but illogical

34 --will explain the function of a program.

Comments

Char **argv can be read as ----

Pointer to pointer to char

What will be output of the following code? String name[4]=

,"ali","bilal","omer",Ayesha"-; Cout<< name [0];

Ali

The data type before a function name represents its,

Return type

37. The expression $((a+b) = 10) \parallel ((x+y) = 20)$ is true only if?

Either $((a+b) =)$ OR $((x + y) = =20)$ is true

45. If $y = 20$ and $z = y ++$, then the value of z will be-----

2146. If

Int a =

17; int

b = 3;

double c = a/b;

what will be value of variable c?

5

What will be the output of the following code?

String grade [3] = {"a", "b", "c"}; Grade[2]

= "f";

Cout<< grade[2];

F

Declaration of two-dimensional array of integers having 3 rows and 5 columns is following?

Int arr [3][5];

A = a + 1; can be written as ----

A+ = 1;

A record is a group of related

Fields

What will be the result of arithmetic expression $6+48/4*3$?

42

Construct is used to execute a set of instructions when a condition is true and other set when it is false?

If\else

If an array has 50elements ,what is allowable range of subscripts?

0 – 49

If `int a = 50;` then the value of `a/= 3;` will be?

16

Which of the following option is/are correct about array?

Only one type of elements can store in an array

Whenever some number is added in an array name , it will jump as many as the added number?

Rows

----- Returns true if c is a digit or a letter and false otherwise.

Int is a l num (int c)

Which of the following is the default mode of the opening using of ofstream class?

ios::trunc

In which language the compiler of c language is written?

C

While loop executes at least?

One time

String are accessed by variable of type----

Char

Which of the following is the correct operator to compare the values of two variables?

= =

Not operator is a -----

Single input operator

Pointer points to a particular-----

Memory address

A function is a block of statements that can be defined on can used-
---in the program.

As many times as user wants

Function is used to find the number of characters in a string?

Strlen()

!=operator is used to check whether the operand on the left-hand side is to the operand on the right-hand-side.

Not equal

A while loop may execute ----- or more times.

0

How we declare an array of integer data whose size is 20 with array name "count".

Int count[20];

A variable declared within a code block becomes ----- variable for that block.

Local

If a file is opened with ios::out mode, then -----

We can write in this file

A function's prototype is written --- the function call.

Before

Loop is mostly used to perform repetition of a tasks for a known number of times.

For

In shift int operations if zero is inserted at the lest most bit, the negative number will become a -----number.

Positive

It is the job of -----to transfer the executable code from hard disk to main memory.

Loader

RAID stands for-----

Redundant array of inexpensive devices

Array indexes start from ----

0

In call by reference, ----- of a variable is passed to the called function.

Address

How many times the following do-while loop will execute?

```
Intk=  
10;  
Do  
{  
Cout << "statements" << endl; ?
```

6

The name of a location in memory is called ----

Variable

Function is used to find the number of character in a string.

Strlen()

Which statement is used to close a file?

Myfile.close();

----- Allow us to have a memory location and use it as int or char interchangeably.

None of the given

While handling files, one can have ----- options

All of the given

----- Function is used to determine the next position to write a character while writing into a file.

Tellp()

Which is the correct syntax to define a structure named 'student'?

```
Struct student{
//variables
};
```

<and> both are ----- operators.

Logical

Identify the logical error in the following if condition 88. If ((i<5) & & (i>15)) ?

Condition is always false

Not operator is a ----

Single input operator

The use of * sign in first and second statements is called --- and ----
of pointer.

```
Int*ptr=
```

```
&x;
```

```
Cout<<*ptr;
```

```
?
```

Referencing, dereferencing

C is widely known as development language of ----- operating system.

Linux

If x is declare as an integer, what will be result of the given expression?

$X = 2 + 3 * 5 \% 2 + 3 / 2$??

4

What will be the correct syntax of declaration of the following statements?

“ptr is a pointer to const int” ??

Const int *ptr

When the logical operator AND(&&) combine two expression exp1 and exp2 then the result will be true only?

When both exp1 and exp2 are true

'multi*3+*3+' element can also be accessed by -----

***(*(multi + 3) + 3)**

What is output of following code?

```
Int data [10] = {10,20,30,40,50,60,70,80,90,100};
```

```
Int*xptr=&data[4]; Int
```

```
*xptr =&data [6];
```

```
Cout<<xptr+yptr++;??
```

120

Which of the following is not the subcategory of the system software?

None of the above

We get $12|8 = \text{-----}$ where "|" is (bit-wise OR)?

12

After evaluating the expression

$X = 10 + 3 * 4 / 2 - 1$, the value of x will be?

15

We can read, write and manipulate the same file using----

Fstream.h

Return type of a function that does not return any value must be----

Void

Pointers store the----

Memory address

-- operator uses the sign^.

Exclusive OR

To get the value store data memory address , we use the----

Dereferencing operator

Telling a computer what to do is called ----

Programming

Compiler translates high level language program into ----- language code.

Machine

Each array declaration must provide the information about all these things except----

The location of the first element to be stored

E of is used to check for the ----- of file when a file is being used.

End

The result of $4\%3$ will be?

1

----- is a substitute of multiple if statement.

If..elseif statement

Find out he logical error in following lines of code. If ($x=10$)

Cout << "x is 10"; ?

Assignment operator should not be used for comparison

Which is not true regarding structure?

we can't declare pointers to structure

114-----are used to compile the code.

Compiler

For converting a void type pointer ptr into integer type, the correct syntax is: **int***

ptr

A pointer with value NULL is defined in the header files _____ and _____.

Stdlib.h stddef.h

Constructor is a special function, called whenever we _____.

Instantiate an object of a class

Constructor is itself a _____ of C++ and _____.

Function, can be overloaded

What is the other name of a variable?

Identifier

In shifting operations if zero is inserted at the left most bit, the negative number will become a _____ number.

Positive

A class is _____.

User defined Data type

Bugs can occur due to _____.

Uninitialized data

The heap memory structure _____.

Constantly changes in size

How many bits from a byte?

8

The function calloc takes two arguments, first argument is the _____ and the second argument is the _____.

Required space in terms of numbers, size of space

The main advantage of function overloading is that _____.

The program becomes more readable

Marco are categorized into _____ type(S)

Two

When we include the header file in the angle brackets the compiler searches for it in _____.

Specific directory

115. #include

<iostream> #include

<cstring> Using

namespace std; Int

main()

{

Charstr1[10]="hello"; Char

str2[10]="world";? **10**

A structure specifies that an action is to be repeated while some

condition remains true.

Repetition

In which language the compiler of c language is written?

C

Characters string are terminated by -----

\0

What will be the value of the variable output in the given piece of code? Double output = 0;

Output = (2 + 2) * 4 + / (4 - 2);

17

What is the output of the following statement? inti = 2.5; do {

couti * 2; } while (i > 3 && i < 10); error

Searching is easier when an array is already sorted True

If an array has 100 elements, what is allowable range of subscripts?

0 - 99

While developing a program; should we think about the user interface?

//handouts main

Yes

A character is stored in the memory in _____

String

These functions seekg() and seekp() requires an argument of type

_____ to let them how many bytes to move forward or backward.

Int

dereferencing operator is represented by ____

A function is a block of statements that can be defined once and used-

-- in the program.

As many times as user wants

Is used for output during the process of writing to the file.

Seekp()

In case of two-dimensional array , if you add some number in a two- dimensional name, it will jump to the next----

Row

In while loop the condition is checked the execution of loop.

Before

Array indexes start from ----

0

Pointers store the----

Memory address

Which step is not performed during a proper program design?

Write details about the performance of program

We can say that a pointer is a variable that holds the-----address.

Memory

When accessing a structure memory, the identifier to the left of the dot operator is the name of ----

Structure variable

From the following; which one is the range of random number generator rand() ?

0 – 32767

To read command-line arguments, the main () function itself must be given -----arguments.

2

Which operator has the highest precedence? (*, -, /, +)?

* and / both have the same highest precedence

Suppose we have `int y[10];`

To access the 4th element of the array we write----

`Y[3];`

transfers the executable code from main memory to hard disk.

Loader

When the logical operator OR (||) combine two expressions `exp1` and `exp2` then the result will be false only, //bitwise "or" applies here....

When both `exp1` and `exp2` are false

suppose we have `int y[10];` To access the 4th element of the array we write_

none of given

we have opened a file stream `myfile` for reading (getting), `myfile.tellg()` gives us the current get position of the file pointer. It returns a whole number of

type_____

int

`ofstream` is used for_____

Output file stream

A structure is a _____ under a single name.
collection of

Variables

In C/C++ the string constant is enclosed ...corrected r red one

In curly braces

The size of `int` data type is 4 bytes

In Flow chart process is represented by

Rectangle

If the break statement is missed in switch statement then, // a break statement with no label always completes abruptly, the reason being a break with no label. If no switch, while, do, or for statement in the immediately enclosing method,

by Attia Kundi

constructor or initializer encloses the break statement, a compile-time error occurs?

This may cause a logical error

When we are using const keyword with a variable x then initializing it at the time of declaration is, ?

Optional

Which of the following is the correct way to assign an integer

value 5 to element of a matrix say 'm' at second row and third

column?

`m[2][3] = 5 ;`

How many total elements must be in two-dimensional array of 3 rows and 2 columns?

6

Which of the following is the correct statement for the following declaration?

`*ptr is a pointer to const int`

What will be the correct syntax to assign an array named arr of 5 elements to a pointer

ptr?

`*ptr = arr[5] ;`

C is a/an language

object oriented

The variables having a name, type and size are just like empty boxes.

True

What's wrong with this for loop? for (int k = 2, k <=12, k++)

the commas should be semicolons

Constructor is a special type of function;

Which has no return type

```
#include <iostream>
#include<conio.h> Using
namespace std;
Void display (int=1, float=2,3, char='@');
Main()
{
Display();
Display(5);
Display(6,7,8);
Getch();
}
Void display( int first float second char third)
{cout<<"intfirst=<<first cout <<second=<<second cout<<third=<<third}
```

First=1, second=2,3, third=@first=5, second=2,3, third =@ first=6, second=7,8 third=@ first=9, second=10,11 third=\$

Default constructor generated by _____does _____for us

Compiler, no initialization

#define CIRCUMFERENCE(X)(2*PI*R) is a

Definition of a marco

Constructor is itself a _____of C++ and ____.

Function, can be overloaded

We cannot use _____ pointer for storing and reading data from it

NULL

Which function is used to delete the allocated memory space?

Free()

the main advantage of function overloading is that _____.

The program becomes more readable

The data members of the class are initialized _____.

At runtime

An instance of a class is called _____.

Object

Malloc function returns _____ to the _____ from the available memory

Void pointer, starting of the chunk of memory allocated from the heap

The _____ is the increment operator in c++

++

The constructor contains _____.

No return type

For accessing data members we use _____ operator

Dot.

In a class we can have _____ constructor

Only one

Object code is machine code but it is not _____ and _____.

Reloadable, executable

Getche() is a _____ function and defined in _____ header file.

Built in function, conio.h

In C++ a variable can be declared anywhere in the program, this will increase _____.

Readability

The compiler use a name mangling technique to _____.

Generate a unique token that is assigned to each function

Most efficient method of dealing with structure variables is to define the structure globally

True

There are mainly _____ types of software.//

Two

When a pointer is incremented, it actually jumps the number of memory addresses ?

According to data type

setw is a parameterized manipulator.

True

if $(a > b \ \&\& \ a > c)$ then the condition will be true
only if Both $a > b$ and $a > c$ are

by Attia Kundi

true

In while loop the loop counter must be initialized,
Before entering the

loop

_____ of a variable means the
locations within a program from where it
can be accessed.

Reference

Switch support data.

Integer Character Float Both integer and character

A function is a block of statements that
can be defined once and used__in the
program.

As many times as user wants

_____ statement is used to
terminate the processing of a
particular case and exit from

switch structure.

break

The most suitable data type for number 325.25 is_.

float

X is an integer variable; what does X++ means,

Add 1 in X value

The operators ++ and -- are used to increment or decrement the value of a variable by_____

1

Which of the following loops checks the test condition at the end of the loop?

Do-While

The operator to free the allocated memory using new operator is

_____.

delete

suppose we have `int y[10]`; To access the 4th element of the array we write_

`y[3];`

From the following; which one is the range of Random number generator function `rand()`?

`0 – 32767`

From following; which one is the correct syntax of an array initialize: Array size is 10 and it is of double data type to value 0?

`double arr[10] = {0.0};`

The string in the array is terminated by a_____

`null`

In C/C++; by default arguments are passed by_____to a function.

`Value`

char name [] = "Hello World"; In the above statement, a memory of characters will be allocated

12

Keyword is used to return some value from a function.

return

Which of the function call is call by value for the following function prototype?

float add(int);

add(x);

The increment of a pointer depends on its ____.

data type

The ASCII code of null character is _____

000

Pointers are a special type of in which a memory address is

store

Transpose of a matrix means that when we interchange rows and columns _____

the first row becomes the first column

Pointers store the
value of a variable

Which of the following function call is “call by reference” for the following function prototype?

int add (int *); add(&x);

The name of the array is a constant pointer which contains the memory address of the of the array.

first element

array name always contains the memory address of the of the array

first element

At the , we try to break up the problem into functional units

design phase

Syntax of a union is identical to _ .

structure

A union is a user-defined data type that contains only _--from its list

of members at a time.

one object

Which of the given file handling modes are similar in their functionality ?

ios::trunc, ios::ate

Which one of the given option is not a mode for reading/writing the data from a file?

Trunk

In C/C++, the header file used for string manipulation is ____.

string.h

A 2D array multi[5][10] can be accessed using the array name as **multi, this technique is called ____.

Double dereferencing

C language was evolved from ____ language(s).

B and BCPL

The ____ statement forces the immediate next iteration of the loop.

Continue

____ are very good tools for code reuse.

Functions

When an argument is passed by reference, ____

the function accesses the argument's original value in the calling program.

____ executes all the lines before error and stops at the line which contains the error.

Intrepreter

Analysis is the _____ step in designing a program.

First

Which of the following is the correct function call, having array named student of 10 elements as a parameter?

addRecord(student);

When an array element is passed to a function, it is passed by

Reference

What is the correct syntax to declare an array of size 10 of int data type?

int name[10] ;

If we want to copy an array, both arrays need to be of same ____.

Data type and size

Which of the following is the correct syntax to print multiple values or variables in a single command using cout?

cout << "H" << x << "\n";

Which one of the statements given below is equivalent to $x \% = 5$?

$x = x \% 5$

The condition in loop should be a(n)_____.

Boolean Expression

Which structure is represented by following flow chart ?

if structure

Stream which sends data to the program is called_____.

Input stream

Syntax of a union is identical to_____.

Structure

When accessing a structure member, the identifier to the left of the dot operator is the name of_____.

a structure variable

The parameter passed to is digit() function is_____.

a character variable

TWAIN stands for_____.

Technology Without An Interesting Name

_____ Translates high level language program into machine language code.

Compiler

From the option given below, what will be the correct output of the code? for (int a = 1; a <= 3; a++)

```
cout<<a*3-1<<" ";
```

2 5 8

If the return type of a function is void then it means that it will_____.

return any type of data return no data

The functions that have already been defined as a part of the language are called _____.

built-in functions

Given that double Array is an array of 7 integers. Which of the below is the correct statement to access the 5th element of the array?

doubleArray[4]

Special name which is substituted in code by its definition and as a result we get an expanded code is called

Macro

$A = a + 1$, can be written as _____

$A += 1$

The compiler generates _____ automatically.

Constructors

The data members of the class are initialized _____

At runtime

- Explicitly write keyword private in the class definition
- Separate the interface and implementation
- The default constructor has no arguments
- Constructor has the same name as of class
- Initializing the data members in the definition of the class is a syntax error

_____ for parameters is also done for inline functions.

Automatic type checking

An instance of a class is called _____.

Object

With the use of dynamic allocation of memory, the system resources can be _____.

Used efficiently

When we include the header file in the angle brackets the compiler searches for it in

Specific directory

___ is a special type of pointer we have to cast it before we use it.

Void

Multiplicative, additive and assignment operators are _____ operators

Binary

The memory allocation functions return a chunk of memory with a pointer of type _____.

Void

Default constructor takes _____.

No parameters

The symbol which is used to represent a process in a flow chart is

Rectangle

When an array is passed to a function , by default it will be passed by_.

Reference

What is the range of Random number generator function rand()?

0 – 32767

The first character of a variable name must be an alphabet or

Underscore

In ___, the first pointer stores the address of the some other pointer, which contains the address of any variable.

single dereference

by Attia Kundi

Initialization of variable at the time of definition is ----- practice.

Good Programming

Which looping process checks the test condition at the end of the loop?

do-while

You can use a _____ statement to transfer control else where out of the nested structure.

Goto

Which of the following array is the most suitable for storing a matrix structure?

Single-dimensional array

The microsoft word document(.doc) is a kind of _____.

Random Access File

A union is a user-defined data type that contains only _____ from its list of members at a time.

one object

In C/C++, _____ are the doors through which data can enter and come out of a program.

Streams

Which one of the given option is not a mode for reading/writing the data from a file?

Trunk

C++ view search file as a sequential stream of _____.

Bytes

In 2D array, the technique used to access the array elements is called

Double dereferencing

From the options given below, identify the correct option which is used for calling the function float square (int &);

square (num);

Identify the correct option which is used for calling the function float area (int).

area(num) ;

Which kind of error occurs in the code below to access the variable num in the main function?

```
int num
=10; int
main(){
int x;
x = num ;
}
```

No error

A program statement that invokes a function is called _____.

function call

What will be the output of the following code segment? main(){

```
int a =
10 ; int
b =0; int
*x=&a;
b =
(*x)+3;
cout << a << " , " << b ;
}
```

1
0
, 1
3

When the logical operator AND (&&) combines two expressions exp1 and exp2 then the result will be true only

When both exp1 and exp2 are true

For which array, the size of the array should be one more than the number of elements in an array?

char

What is the range of Random number generator function rand()?

0 – 32767

To assign a value to a character type variable, _____ are used around the value.

single quotes ' '

In C/C++, the #include is called _____.

Preprocessor Directive

The number 544.53 must be stored in _____ data type.

Float

Use of the _____ statement violates the rules of structured programming.

Goto

In Flow Chart, flow of control is represented by _____.

Arrow

What will be the correct syntax for initialization of pointer ptr of type int with variable x?

int *ptr = &x ;

Syntax of a union is identical to _____.

structure

A union is a user-defined data type that contains only _____ from its list of members at a time.

one object

Which of the given file handling modes are similar in their functionality ?

ios::trunc, ios::ate

Which one of the given option is not a mode for reading/writing the data from a file?

Trunk

Stream which sends data to the program is called _____.

Input stream

Syntax of a union is identical to _____.

Structure

When accessing a structure member, the identifier to the left of the dot operator is the name of ____.

a structure variable

The parameter passed to isdigit() function is _____.

a character variable

TWAIN stands for _____.

Technology Without An Interesting Name

_____ Translates high level language program into machine language code.

Compiler

From the option given below, what will be the correct output of the code? for (int a = 1; a <= 3; a++)

```
cout<<a*3-1<<" "; 2
```

5 8

If the return type of a function is void then it means that it will _____.

return no data

The functions that have already been defined as a part of the language are called _____.

built-in functions

Given that doubleArray is an array of 7 integers. Which of the below is the correct statement to access the 5th element of the array?

doubleArray[4]

The symbol which is used to represent a process in a flow chart is _____.

Rectangle

When an array is passed to a function, by default it will be passed by ____.

Reference

The first character of a variable name must be an alphabet or _____.

Underscore

In _____, the first pointer stores the address of the some other pointer, which contains the address of any variable.

single dereference

Initialization of variable at the time of definition is ----- practice.

Good Programming

Which looping process checks the test condition at the end of the loop?

do-while

You can use a _____ statement to transfer control else where out of the nested structure.

Goto

An object of _____ class can be created for saving the current position of any opened file

before reading from the file Stream

The value returned by the of() function is of _____ type.

Character

If a file is not properly closed in a program then what is the status of the program ?

The program will terminate normally

The parameter passed to is digit() function is _____.

a character variable

In C/C++, the header file used for string manipulation is _____.
string.h

Arrays are best data structures _____.
for relatively permanent collections of data

Each array declaration must provide the information about all these things except _____.
Number of elements of the array

The First step for developing a program is _____.
Linking Analysis

_____ Transfers the executable code from hard disk to main memory.
Loader

_____ Operator Is used to pass the address of a variable to the called function.
&

In call by reference, _____ of a variable is passed to the called
function. Address

_____ of a function is also known as signature of a function.

Declaration

Preprocessor or program performs its function before _____ phase takes place. **Compiling**

Which of the following is the correct function call, having array named student of 10 elements as a parameter?

addRecord(student[10]);

The remainder (%) operator is a _____ operator.

Arithmetic

Which of the following array is the most suitable for storing a matrix structure?

Two-dimensional array

*.doc is _____ by type.

Random Access File

Which of the following is NOT a preprocessor directive?

#define

The return type of operator function must always be void.

False

What does (*this) represent?

A value of the data member

The statement cin.get(); is used to,

Read a character from keyboard

When we do dynamic memory allocation in the constructor of a class, then it is necessary to provide a destructor.

True

Overloaded new operator function takes parameter of type size_t and returns

void pointer

These are the parameters of operator functions for << and >> are objects of the class for which we are overloading these operators.

True

C++ is a case-sensitive language

True

To include code from the library in the program, such as `iostream`, a directive would be called up using this command.

#include <iostream.h>

A template function must have only generic data types.

True

Template class can not have static variables.

False

What will be the correct syntax to assign an array name `arr` of 5 elements to a pointer `ptr`?

***ptr = arr[5] ;**

What will be the correct syntax to access the value of fourth element of an array using pointer `ptr`?

Both 1 and 3

If most significant bit of un-signed number is 1 then it represents a positive number.

True

If there is a symbol (& sign) used with the variable name followed by data type then it

Refers to__and if & is being used with variable name then it refers to

____.
Reference variable, address of variable

We can also do conditional compilation with preprocessor directives.

True

The default value of a parameter can be provided inside the

function definition

Classes defined inside other classes are called_____class

nested

What purpose do classes serve?

All of the given options

Every class contains_____.

Constructor

New operator is used to allocate memory from the free store during

None of the given options

When an object of a class is defined inside another class then,

Destructor of inner object will be called first

It is possible to define a class within another class.

True

New and Delete are also used with _____ and data types as well.

Class, Objects

With New keyword, data types and class members are initialized with meaningful values instead of garbage.

True

C language is developed by

Dennis Ritchie

Which of the following choice is not an example of an int data type?

-4.0

In flow chart, the symbol used for decision making is,

Diamond

Switch statement deals with,

Integer and character data

Default case in switch statement is,

Optional

*.doc is _____ by type.

Sequential File

Member function tell g() returns the current location of the ____

pointer.

tellptr()

Dealing with structures and functions passing by reference is the most economical method

True

In C/C++ all character strings are terminated with,

Null character

Word processor is <http://vustudents.ning.com>

Application software

Which of the following can not be a variable name?

10area

Which looping process is best, when the number of iterations is known?

while

By default an array of characters is passed by value to a function

False

Which of the following operator is used to access the address of a variable?

& operator

The name of an array represents address of first location of array element.

False

Compiler is a

Application Software

If Num is an integer variable then Num++ means,

Add 1 with Num

For one byte there are _____ combinations of values that can be stored in computer.

 2^8

In C/C++ language the header file which is used to perform useful task and manipulation of character data is **ctype.h**

Default case in switch statement is,

Optional

When break statement is encountered in switch statement, it

Exits from switch statement

will be the result of arithmetic expression $6 + 27/3 * 3$?

What is the correct syntax to declare an array of size 10 of int data type?

int name[10] ;

How many dimensions does n-dimensional array has?

n dimensions

What will be the correct syntax to access the value of fourth element of an array using pointer ptr?

ptr[3]

Which of the following values C++ use to represent true and false?

1 and 0

Declaring structures does not mean that memory is allocated.

True

For which array, the size of the array should be one more than the number of elements in an array?

char

In a class we can have _____ constructor(s).

Only one

Which one of the following is mandatory preprocessor directive for c++?

#include <isotream>

Header files provide _____ so the program running on one operating system can run without an error on the other system

Accessibility

We should not use such variable names that are starting with _____ because in C++ these are lots of internal constants and symbolic names that start with it.

Double underscore

Which of the following code segment represents a left shift operator?

Result= number<<1

Which of the following code segment represent a bitwise And operator?

&

Which of the following code segment represent a bitwise OR operator?

|

_____ of the following code segment represent a Bitwise Exclusive OR operator

Which of the following code segment represent a NOT operator

~

If a variable is passed by value to a function and the function makes some changes to that variable then it

does not affect the original variable

In C/C++ the #include is called,

Preprocessor Directive

Loops are _____ Structure.

Repetition

The size of operator is used to determine the size of _____.

data type

Structures use _____ allocation.

Stack

Which header file must be included to use the functions to lower() and to upper()?

ctype.h

In Analysis, we try to have a _____

Precise problem statement

CS201

Mid-Term

Quiz by Attiq Kundi

SUBSCRIBE "Knowledge For Us" ON YOU TUBE

100 % CORRECT MCSQ
AND Many MoRe

CS 201

by Attia Kundi

Pointers store the _____

memory address

_____Returns true if c is a digit and false otherwise.

int isdigit(int c)

e of is used to check for the _____ of file when a file is being read.

End

The statement cout<<ypr will show the _____ the y ptr points to.

memory address

Which of the following is true about streams?

A& B

While handling files ,one can have _____options

All of the given

If an array has 50 elements, what is allowable range of subscripts?

0 – 49

A structure is a collection of _____ under a single name.

variables

If stream is used for _____

Input file stream

The ASCII code of null character is _____

000

dereferencing operator is represented by _____

When ever some number is added In an array name , it will jump as many _____ as the added number.

rows

To access the data members of structure, _____ is used.

Dot operator

Which of the following is true about streams?

A. It is a sequence of bytes

B. It is an ordered sequence

C. All bytes can go through the stream simultaneously

D. Bytes that enters first into the stream will go out at last

A and B

We use _____ to open a file or stream for extraction

in

char**argv can be read as _____.

pointer to pointer to char

From following; which one is the correct syntax of an array initialize: Array size is 10 and it is of double data type to value 0?

double arr[10] = {0.0};

Pointers work by pointing to a particular _____

data type

_____ function give the position of the next character to be read from that file.

tellg()

aFile. seekg (-10L, ios:cur)

backward direction

An object of _____ class can be created for saving the current position of any opened file before reading from the file.

Stream

Structures are syntactically defined with the word _____.

struct

_____ Allow us to have a memory location and use it as intorchar interchangeably

union

A hierarchy of classes which are used to deal with console and disk files are called _____.

IO classes

_____ function is used to determine the next position to write a character while writing into a file.

tellp()

For binary member operators, operands on the _____ drives(calls) the operation.

Left

We cannot increment _____.

references

We can _____ pointer.

all of the given

We can _____ references.

None of the given

What will be the correct syntax for the following function call?

add(x);

An instance of a class is called _____.

object

The _____ is called automatically when an object destroys

destructor

The destructor is used to _____.

deallocate memory

_____ data isn't accessible by non-member functions or outside classes.

private

Member functions of the class _____ main program.

are accessible from

Overloading means :

Using the same name to perform multiple tasks or different tasks depending on the situation.

The main advantage of function overloading is _____.

The program becomes more readable

You cannot overload the _____ operator.

? :

In C++, a variable can be declared anywhere in the program this will increase ____.

efficiency

Memory allocated from heap or free store _____.

cannot be returned back unless freed explicitly using free and delete operators

We cannot use _____ pointer for storing and reading data from it.

'NULL

The dynamic memory allocation uses _____ where as static memory allocation uses ____.

heap , stack

What will be the output of the given code? #include

```
#define MAX( A,B)((A)>(B)?(A):(B)) void main() {
    int i,
    x,y;
    x =
    23;
    y =45;
    i = MAX( x++, y++ );
    // Side-effect: // larger value incremented twice cout << "x = " << x << " y
    = " << y << '\n';
}
```

x=24 y=47

NULL has been defined in _____ header file.

Stdlib.h

Symbolic constant PI can be defined as:

#define PI 3.14

The friend function of a class can have access _____.

to the private data members

C++ was developed by _____.

Bejarne Stroustrup

Once the _____ are created, they exist for the lifetime of the program.

static variables

Encapsulation means _____.

that the data of a class can be accessed from outside

An address is a _____, while a pointer is a _____.

constant, variable

The syntax of declaration of a function that returns the reference to an integer is _____.

int & myfunc();

Which one of the following is mandatory preprocessor directive for c++?

#include

The members of a class declared with the key word structure _____ by default.

Public

getche() is a _____ function and defined in _____ header file.

built-in function , conio.h

_____ operators are the ones that require two operands on both sides of the operator.

Binary

_____ Will return the number of bytes reserved for a variable or data type.

Size of operator

_____are not available in C language.

Library functions

The members of a class declared without any key word are _____ by default .

private

For console input and output we use _____.

conio.h header file

The name of the destructor is the same as that of a class proceeding with a ____.

~ sign

A reference cannot be NULL it has to point a data type.

True

A pointer is_____.

a variable for storing address

Constructor is a special function, called when ever we _____.

instantiate an object of a class

Symbolic constant PI can be defined as:

by Attia Kundi

#define PI 3.14

Object code is machine code but it is not _____ and _____.

relocatable, executable

The default visibility for the data members of the class is private

accessible outside the class

The _____ is called automatically when an object destroys.

destructor

Constructor is special type of function : which has no

return type

_____ variables are those that are defined outside of main.

Global

With in the statement obj1=obj2;obj1 will call the assignment operator function and obj2 will be passed as an argument to function.

True

When the compiler overload the assignment(=)operator by default then

Compiler does member wise assignment.

It is possible to return an object from function using this pointer.

True

Overloaded assignment operator must be Member function of class

Let suppose

Int a, b, c, d, e;

a = b = c = d = e = 42;

This can be interpreted by the compiler as a=(b=(c
=(d= (e = 42)))));

In statement a+b+c, at first b+c is
executed first

Suppose int i = 10; then what is the output of cout<<oct<<i;

12

O stream is a _____ operator.

Standalone

_____ Must be included to use stream manipulation in your code.

Io manip

_____ Operators are the ones that require only one operator to work.

Unary

The end land flus hare _____.

Manipulators

When operator function is implemented as member function then return type of function ____.

Can be any data type

by Attia Kundi

When a variable is defined as static in a class then_____.

Only one copy is created for all objects of this class

Automatic variables are created on_____.

Stack

```
cout<<i<<"";
```

```
cout<<d<<"";
```

```
cout <<f;
```

Above statements can be written within statement of one line as:

cout << i << " " << d << " " << f;

dec,hex,octareall_____.

Non-parameterized manipulators

What will be the output of following statement?

```
cout<<setfill('0')<<setw(7)<<128;
```

0000128

Which of the following syntax is best used to delete an array of 5objects named 'string' allocated using new operator.

delete []string;

If we have a program that writes the output data(numbers) to the disc, and if we collect the output data and write it on the disc in one write operation instead of writing the numbers one by one.

In the above situation the area where we will gather the number is called Buffer

The first parameter of operator function for << operator_____.

Must be passed by reference

The second parameter of operator function for >> operator must always be passed

By reference

The only operator that the compiler overloads for user define data type by default is-----

Assignment (=) operator

Consider the following code, the printed value will be converted into: int n=10;

cout << oct << n;

Base 8

_____ variables are defined in the main.

Local

O stream class is _____ and not under our control.

built-in

The memory allocation in C++ is carried out with the help of _____.

new operator

If B is designated as friend of A, B can access A's non-public members.

A can access non-public members of B

If the request of new operator is not fulfilled due to insufficient memory in the heap _.

the operator returns 0

We should not use such variable names that are starting with _____ because in C++, there are lots of internal constants and symbolic names that start with it.

double underscore

The friend key word provides access_____.

in one direction only

The malloc function takes_____argument(s).

one

The constructor contains_____.

no return type

What will be the output of the following c++ code?

```
#include<iostream.h>
#define max 100
main()
{
#ifdef max
Cout<<"Hellow;
}
```

Error

Once we have defined as symbolic constant valueu sing#define, that value

during program execution cannot be changed

The memory allocation functions return a chunk of memory with a pointer of type _.

ptr

A class can be declared as a _____ of other class.

member

To avoid dangling reference, don't return _____.

the reference of a local variable from the function

Constructor is its elfa _____ of C++ and _____.

function, can be overloaded

The parameter passed to isdigit() function is _____ variable.

Integer

char**argv can be read as _____.

pointer to pointer to char

To read command-line arguments, the main() function itself must be given _____ arguments.

2

How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed?

```
intPtr += 2;
```

8

The increment of a pointer depends on its _____.

data type

The statement `cout<< yptr` will show the _____ the `yptr` points to.

memory address

_____ is used as a dereferencing operator.

Transpose of a matrix means that when we interchange rows and columns _____.

the first row becomes the first column

Individual characters in a string stored in an array can be accessed directly using array_____.

subscript

We can define a matrix as _____ array.

Multi dimensional

A _____ is an array of characters that can store number of character

specified. String

Given a two dimensional array of integers, what would be the correct way of assigning the value 6 to the element at third row and fourth column?

array[2][3] = 6;

_____ of a variable means the locations within a program from where it can be accessed.

Visibility

Which of the following function call is "call by reference" for the following function prototype?

int add (int *); add(&x);

Which of the following function call is "call by reference" for the following function prototype?

float add (float *); add(&x);

Which of the function call is call by value for the following function prototype?

float add(float); add(x);

by Attia Kundi

Which of the function call is "call by value" for the following function prototype?

float add(int);

add(x);

Return type of a function that does not return any value must be _____.

void

_____ will be used for enclosing function statements into a block.

{ }

What is the output of the following code if the 2nd case is true switch (var){

case 'a': cout<<"apple"<<endl; case

'b':cout<<"banana"<<endl; case

'm':cout<<"mango"<<endl; default: cout<<"any

fruit"<<endl;

}

Banana mango any fruit

When the break statement is encountered in a loop's body, it transfers the control from the current loop.

Outside

What is the output of the following code if the 3rd case is true

switch (var) {

case 'a':cout<<"apple"<<endl; case

'b':cout<<"banana"<<endl; case

'm':cout<<"mango"<<endl; default: cout<<"any

fruit"<<endl;

}

Mango any fruit

What is the output of the following code, if the first case is true

```
switch (var){ case 'a':cout<<"apple"<<endl; case
'b':cout<<"banana"<<endl; case
'm':cout<<"mango"<<endl; default: cout<<"any
fruit"<<endl;
}
```

Apple banana mango any fruit

What will be the output of following code segment? for(int i=2; i<10;

```
i++){
if(i==5)continue;
cout<<i<<" ";
}
```

2,3,4,6,7,8,9

Statement is used to terminate the processing of a particular case and exit from switch structure.

break

What will be the result of the expression $j = i++$; if initially $j = 0$ and $i = 5$?

5

What will be the result of the expression $k = ++m$; if initially $k = 0$ and $m = 5$?

6

How many times the following do-while loop will execute?

```
int k=10; do{cout<<"Statements"<<endl; k-=2;}while(k>0);
```

6

Which of the following loops checks the test condition at the end of the loop?

by Attia Kundi

Do-While

The operators++and— are used to increment or decrement the value of a variable by _.

1

How many times the following loop will execute?

```
int j=3; while(j>0){cout<<"Statements"<<endl; j--;}

```

2

A ____ structure specifies that an action is to be repeated while some condition remains true.

Control

!(x>3) means in C++ that x is less

than or equal to 3

When the logical operator && combines two expressions then the result will be true only when the both expressions are _____.

true

<and> both are _____ operators.

Relational

What will be the value of variable “input” if the initial value of input is 67?

```
if(input >= 50)
    input = input + 1;
if(input <= 75)
    input = input + 2;

```

else

input = input - 1;

70

!(x < 3) means in C++ that

x is greater than or equal to 3

!=operator is used to check whether the operand on the left-hand-side is _____ to the operand on the right-hand-side.

Not equal

When the if statement consists more than one statement then enclosing these statement in curly braces is,

Must

The most suitable data type for number 325.25 is _____.

Float

What will be the result of arithmetic expression $6 + 48 / 4 * 3$?

42

Which of the following will be the most appropriate data type to store the value 63.547?

Float

In the given expression which operator will be evaluated first? $10 + (6 / 2) - 2 * 3$?

/

What will be the value of the variable output in the given piece of code? double output

= 0; output = $(2 + 2) * 4 + 2 / (4 - 2)$;

17

It is the job of _____ to transfer the executable code from hard disk to
main memory. Loader

In computer systems the reared mainly _____ type of softwares .

2

_____ will explain the function of a program.

Comments

if(a>b&&a>c) then the condition will be true only if Both a>b and a>c are
true

A variable of character data type occupies _____ byte(s) in memory.

1

We must include the header file _____ to convert the value of one
 type into another type using built-in functions

. string.h

A function is a block of statements that can be defined once and used
 _____ in the program.

As many times as user wants

Select the correct way to assign the address of first element of array to pointer?

int *ptr = &data;

Consider the following code segment. What will be the output of following code?

```

int addValue (int *a){ int b
= (*a) + 2; return b;
}

main()
{ int x
= 6;
cout<<addValue(&x)<<" "; cout<<x;
}

```

6,8,6

Here the code is given below. You have to identify the problem in the code. while(i < 10) && (i > 24))
the test condition is always false

The correct syntax of do-while loop is _____.
do {statements;} while (condition);

Matrix is defined as _____.
Multi-dimensional array

In programming, comments are used to explain the functioning of the _____.
Program

Operating System is a type of a/an _____.
system software

From the options given, you need to choose the option which is true for the given code.

```
for (int i = 1; i>0; i++) {  
    /*loop code*/  
}
```

the test condition is always true

by Attia Kundi

Which of the following values are used in C/C++ to represent true and false?

1 and 0

'While' loop may execute _____ or more times.

zero

Body of any function is enclosed within _____.

{ }

What will be the correct syntax for initialization of a pointer ptr with string "programming"?

char *ptr = "programming";

Which one of the given option is not a mode for reading/writing the data from a file?

Get

Which of the following operators is used to access the value of variable pointed by a pointer?

*** operator**

In case of singled referencing, the value of the _____ is the address of the__.

pointer, variable

The remainder (%) operator is a _____ operator.

Arithmetic

What will be the output of following code? int x=10;

cout<<"x="<<x;

x=10

The purpose of using `cout << is to _____`.

Display information on the screen

Which of the following data types will be assumed if no data type is specified with constant?

Short

All the preprocessors directive starts with _____.

#

NULL has been defined in _____ header file.

stdlib.h

Which one is the correct example of a unary operator

&

In which statement there can be _____ default statement(s)

1

To access the data members of structure _____ is used

Dot operator

The statement `cout << yptr` will show the _____ the yptr points to

Memory address

If an array has 50 elements. What is allowable range of subscript

0-49

A^b can be written as _____

$A=a^b$;

If $j=10$ and $x=++j$ then the value of x will be _____.

11

The string in the array is terminated by a_____.

Null

In C++, what will be the highest index of the following array? `Int array{15}`

14

Which of the following is a correct way to assign the address of the first element of array to pointer? `Int data[10];`

`Int*ptr= data [0]`

For casting, we normally declare a pointer of type_____.

We are going to use

`A=aIb;` can be written as_____.

`aI=b;`

The exclusive OR operator returns 1 if_____.

One inputs is 1 and other is zero

Sometimes after allocating memory we need additional space, for this purpose we use

_____.

Reallocate function

An integer uses four bytes and the integer calculations occur in_____bytes.

4

Shifting the binary numbers is similar to shifting the_____numbers

Decimal

_____operator uses the sign^

Exclusive OR

When an array element is passed to a function, it is passed by_____.

reference

While programming, it is good to provide an easy to understand and easy to use interface; this programming skill is called_____.

usability

_____executes all the lines before error and stops at the line which contains the error.

Intrepreter

Which of the following is the correct syntax to access the value of first element of an array using pointer ptr?

***ptr[0]**

C is a/an_____language.

function oriented

_____of a function is also known as sig nature of a function.

Declaration

_____Are very good tools for coder use.

functions

If any break statement is missed in switch statement then_____.

this may cause a logical error

A 2D array multi[5][10] can be accessed using the array name as **multi, this technique is called_.

Double dereferencing

In C/C++, the default command line arguments passed to the main function are _.

int argc, char **argv

Record is a group of related_____.

Fields

The Microsoft Word document(.doc) is a kind of_____.

Random Access File

NULL character is used to indicate the_____of string.

End

How many dimensions does n-dimensional array have?

n dimensions

Which of the following function call is "call by reference" for the following function prototype?

func(&num);

The loop which is most suitable to be used when the number of iterations is known is called _.

for

In C/C++, the string constant is enclosed in_____.

double quotes " "

by Attia Kundi

In order to get the right most digit of a number, we divide this number by 10 and take__.

Its remainder

What is the correct syntax to declare an array of size 10 of int data type?

int name[10];

How many bytes of memory are occupied by array 'str'? char str[]="programming";

11

Suppose that an integer type pointer contain same memory address 0x22f230. What will be the new memory address if we increment this pointer by one?

0x22f234

Which of the following if missing would result in infinite recursion in case of recursive function?

Recursive call

When ever we use a library function or a predefined object or macro, we need to use a __.

header file

Switch statement deals with _____ type of data.

Both Integer and Character

Both compiler and _____ are used to translate program into machine language code.

Interpreter

TWAIN stands for _____.

Technology Without An Interesting Name

The parameter passed to is digit() function is _____.

a character variable

C++ view search file as a sequential stream of _____.

Bytes

Structure is a collection of _____ under a single name.

only variables

The default mode for writing into a file using of stream object is _____.

out

The memory address of the first element of an array is called _____.

base address

We want to access array in random order which of the following approach is better?

Pointer

The _____ structure is a multiple-selection construct which makes the code more efficient and easy to read and understand.

switch

Which of the following is not a reserved word in C/C++?

Sum

To access rand(), which library is required to be included in program?

stdlib.h

What is the highest legal index for the following array?

int arr[4]

3

Word process or is a type of a/an _____.

application software

Identify the correct option which is used for calling the function float area (int).

area(int num);

The _____ statement allows us to select from multiple choices based on a set of fixed values for a given expression.

switch

C is widely known as development language of _____ operating system.

Unix

To convert the value of one type into another type using built-in functions, we include _____ header file.

stdlib.h

The keyword _____ is used to get some value back from a function.

return

The function seek g() takes _____ parameter(s).

2

The function write() takes _____ as parameter(s).

String and no. of bytes to be written

When the logical operator AND (&&) combines two expressions exp1 and exp2 then the result will be true only _____.

When both exp1 and exp2 are true

Syntax of a union is identical to _____.
structure

In Flow Chart, flow of control is represented by _____.
Arrow

There can be _____ 'default' statement(s) in any switch structure.
2

The condition in loop should be a(n) _____.
Arithmetic Expression

How many nested loops would be required to manipulate n-dimensional array?
2n

Which of the following is not an example of int data type?
-4

We should use _____ for clarity and to force the order of evaluation in an expression.
parenthesis ()

Which of the following is the starting index of an array in C++?
0

The statement $x+=y$ can be interpreted as _____.
Adding the value of the x to the value of the y and storing the result in x

Given a 2D array of integers, what would be the correct way of assigning the value 5 to the element at second row and third column?

m[1][2] = 5;

Array is a data structure that stores _____.

Variables

A program statement that invoke as a function is called _____.

function call

The compiler of C language is written in _____ language.

C

Which function is used to locate the first occurrence of a character in any string?

strlen()

To access the data members of structure, _____ is used.

Dot operator

In the following nested For Loop, which loop will run most number of times? for(int i = 0; i < 5; i++)

{

for(int k = 0; k < 5; k++)

{

.....

}

}

Depends upon the statements in the inner loop's body

Structure use _____ allocation.

Heap

Function give the position of the next character to be read from that file.

tellp()

What will be the size of the following character array? `char name[] = "Adeel";`

5

Function prototype is written,

Before call of that function

Which one of the following languages has been used to write the compiler of "C" language?

C

A hierarchy of classes which are used to deal with console and disk files are called ____.

Stream classes

_____Stops execution at the line that contains error(s)In the code.

Interpreter

C++ is a _____language.

High level

How many elements are stored in the following? `int matrix [4][5];`

20

_____is a substitute of multiple if statement.

if. .else if statement

if

int sum = 54;

Then the value of the following statement is sum = sum -3;

51

What will be the correct syntax for declaration of the following statement? "ptr is a constant pointer to an integer"

const int *myptr;

_____ operator is used to pass the address of a variable in call by reference method.

&

_____ data type can operate on modulus operator.

int

When ever some number is added in an array name, it will jump as many _____ as the added number.

rows

Suppose that an integer type pointer contain same mory address 0x22f220. What will be the new memory address if we increment this pointer by one?

0x22f224

_____ Is the pointer which determines the position in a file from where the next read operation occurs.

put

what is the correct syntax to declare an array of 15 elements of type float?

float array [14]

suppose we have a file with name "mylife.txt". what will be the correct Syntax of file opening for input?

my file (—myfile.txt||. ios::in)

Which of the following will be the most appropriated at a type to store the value 63.5 47?

float

-----will explain the function of a program

Comments

These functions Seek g()and seek p() requires an argument of type-----

-- to let them how many bites to move forward or backward Int

All elements of an array must be of----- data type

Same

what will be the result of arithmetic expression $6 + 48 / 4 * 3$?

42

-----allow us to have a memory location and use it as intor char interchangeably

none of the given

accessing a structure member the identifier to the left of the dot operator is the name of-----

structural variable

<and> both are ----- operators

Arithmetic

what will be the output of the following code segment? For(int=2;i<10;i++){

If (i == 5)

Continue;

cout<<i<<“.”;

}

2,3,4,5,6,7,8,9

----- returns true if she is a digit or a letter and false otherwise

Int isalnum (int c)

If a --- find an error, it stop translating without generating any E executable code

Compiler

the ASCII code of null character is-----

000

If X=9; then the expression sx%=2; will calculate the value of x as-----

- 1

the original value means unchanged in -----

Call by Value

which of the following is used to create an account stream?

Ofstream

how many times the following loop will execute? Int j = 3;

While (j > 0)

{

Cout << "statements" << endl;

J - = 2;

}

2

Use of word processor for writing the code of a program is not good because -----

It has less feature for writing code of a program

how many bytes and integer type pointer int ptr will jump in memory if the statement below is executed?

Int ptr

+=2;

8

Avoid loop may execute ----- or more Times

0

raid stands for-----

redundant array of inexpensive dices

in a file is opened with iOS:: out mode, then-----

a new file is created

first step in design in gaprogram is **Analysis**

what will be the result of the expression $J = i++$

If initially $j=0$ and $i=5$? 5

suppose we have `int y [10];`

to access the 4th element of the array be write-----

`Y [3];`

A -----structure specify that an action is to be repeated while

some condition remains true Control

the most suitable data type for number 325.25 is-----

float

Which one of the following should be placed immediately either before or after the noun?

Modifiers of Nouns

Titles typically include both.....

a number and a description

For what purposes colons are used?

All of the above

The name of the array is a constant pointer which contains the memory address of the of the array.

first element

What will be the correct syntax for the following function call?

float add (int &); add(x);

In C/C++; by default arguments are passed by _____ to a function.

Value

<, <=, >, >= are called _____ operators.

Relational

char name [] = "Hello World" ; In the above statement, a memory of _____ characters will be allocated

12

While developing a program; should we think about the reusability?

Yes

C is widely known as development language of _____ operating system.

Unix

Which of the following function call is "call by reference" for the following function prototype?

int add (int *); add(&x);

If a function has not been declared before its definition,

It is a None of these

While loop executes at least----

Zero time

A _____ is a precise sequence of steps to solve a particular problem. **Program**

In while loop the loop counter must be initialized,
Before entering the loop

```
int counter = 0 ; while(counter < 15) cout << counter << " "; counter++;
```

Infinite loop

Why we use "cin"?

To read data from keyboard

Loops are _____ Structure
Repetition

Topic 13 to 18

By

Attiq Kundi

In an n-dimensional array , we use _____ for loop(s) to populate the

array n

In 3 dimensional array, we use _____ for loop(s) to populate the array

3

In 2 dimensional array, we use _____ for loop(s) to populate the array

2

In 1 dimensional array, we use _____ for loop(s) to populate the array

1

A group is a record of related _____

Fields

__contains functions for manipulations of

character data ctype.h

Ifstream inFile refers _____

Object for reading from a file

ofstream outFile refers _____

object for writing to a file

A character is stored in the memory in _____

Byte

A string which is assigned to a string variable is terminated by

Null character

When an identifier is declared with keyword const

then, Its value cannot be changed

If stream is used for _____

Input file stream

Which of the following option is correct about array

Only one type of elements can be stored in an array

Suppose that an integer type pointer contains a memory address 0x221220. What will be the new memory address if we increment this pointer by one

0x221224

In the case of pointer to pointer or _____ the first pointer contains the address of the second pointer, which contains the address of the variable, which contains the desired value

Double dereference

The data in the memory is _____

Volatile

Suppose we have a file with name myfile.text What will be the correct syntax of file opening for input

myFile,open("myfile.text", ios::in);

In analysis, we try to have a _____

Precise problem statement

__Returns true if c is a digit or a letter and false

otherwise int isalnum (int c)

____Returns true if c is a digit and false

otherwise int isdigit(int c)

____Returns true if c is a letter and false

otherwise Int isalpha(int c)

__Returns true if c is a hexadecimal digit character and false otherwise

Int isxdigit(int c)

____Returns true if c is a lowercase letter and false

otherwise Int islower(int c)

__Returns true if c is an uppercase letter and false otherwise

Int isupper(int c)

__if c is an uppercase letter, tolower returns c as a lowercase letter. Otherwise, tolower returns the argument unchanged

Int tolower (int c)

__if c is a lowercase letter, toupper returns c as an uppercase letter. Otherwise, toupper returns the argument unchanged

Int toupper(int c)

Returns true if c is a white-space character newline ('\n'), space (' '), form feed ('\f'),

carriage return ('r'), horizontal tab ('\t'), or vertical tab ('\v') and false otherwise

Int isspace(int c)

__ returns true if c is a control character and false

otherwise Int iscntrl(int c)

_____ Returns true if c is a printing character other than a space, a digit, or a letter and false otherwise

Int ispunct (int c)

Return true value if c is a printing character including space (' ') and false otherwise

Int isprint (int c)

Returns true if c is a printing character other than space(' ') and false

otherwise Int isgraph(int c)

The function tolower and toupper are _____

Conversion function

Which header file must be included to use the function tolower() and toupper()

Ctype.h

Which mode of file is used to open a file or stream for

insertion Out

Which mode of file is used to open a file or stream for extraction

In

Append rather than truncate an existing file. Each insertion will be written to the end of the file

App

Discards the file's contents if it

exists. Trunk

Opens the file without truncating, but allows data to be written anywhere

in the file Ate

Treat the file as binary rather than text. A binary file has data stored in internal formats, rather than readable text format

Binary

Following is the declaration of a _____ array. Int

arr[2][2] 2 dimensional

____ are conventional names of the command line parameters of the 'main()' function.

'argc' and 'argv'

What will be the correct syntax for declaration of the following statement

'ptr is a constant pointer to an integer

Int*const ptr;

What will be the correct syntax for declaration of the following statement

'ptr is a pointer to const in'

`const int *ptr.`

Transpose of a matrix means that when we interchange rows and columns_____

The first row becomes the first columns

Char dest[50] = using strcat function”;

Char src[50] = this part is added and this is ignored;

Strncat(dest, src, 19);

Cout <<dest;

Using strcat function , this part is added

Base address is the memory address of _____element of an array.

1st

While handling files, one can have _____options

Read the file, write in some file, read and write in the some file...all

String are accessed by variable of type

Char

For the following statement, how much memory will be allocated to my array[0]

Char*myarray[] = {“arif”,”sadia”}

5 bytes

To get the value stored of a memory address we use the _____

Deferencing operator

Data means _____

The value of whatever the data points to

If a file is opened ios::out, then _____

A new file is created

If we want to append into the file, the mode will be _____

ios::app

If we want to write anywhere in the file, the mode is _____

ios::app

Following is the declaration of a 2D array, how many row are declared for this array `int arr[3][2]`

3

Following is the declaration of a 2D array, how many columns are declared for this array `int arr[3][2]`

2

In case of two dimensional array, if you add some number in a two-dimensional array name, it will jump to the next _____.

Rows

Which of the following is true about streams? It is a sequence of bytes

It is an ordered sequence

All bytes can go through the stream simultaneously Bytes
that enters first into the stream will go out at last

A and B

Which of the following is a correct way to assign the address of the first element of array is pointer? `Int data[10]`

`Int*ptr=data[0];`

`A_` is an array of characters that can store number of character
specified String

A string which is assigned to a string variable is terminated by_____.

Null character

How many bytes an integer type pointer `int ptr` will jump in memory if the statement below is executed? `Int ptr +=2`

8

Give a two dimensional array of integers, what would be the correct way of assigning the value 6 to the element at third row and fourth columns

`Array[2][3]=6;`

Following is an array of _____rows and _____, int

`num[3][4];` 3 and 4

What will be the output of the following code segment?

`Char myName[]="student ID";`

`Int n=sizeof(myName);`

`Cout<<n;`

11

If we have a pointer on the left hand side then the right hand side should have

a/an _____

Address

The parameter passed to isdigit() function is _____ variable

Character or integer

Ofstream is used for _____

Output file stream

A hierarchy of classes which are used to deal with console and disk files are called _____

Stream classes

'\0' takes _ space in the memory.

1 byte... not sure

Which statement is used to close a file

Myfile.close()

Which statement is used to open a file

My file.open()

We must include the header file _____ to convert the value of one type into another type using built-in-function.

String.h

Pointer points to a particular _____.

Memory address

Ifstream is used for _____

Input file stream

Ofstream is used for

Output file stream

To get the value stored at a memory address, we use the _____.

Deferencing operator

To read command-line arguments, the main() function itself must be given _____ arguments.

2

When an identifier is declared with keyword const then,

Its value cannot be changed

Constructor is itself _____ of C++ and _____

Function, can be overloaded

In analysis, we try to have a _____

Precise problem statement

In c++ a variable can be declare anywhere in the program this will

increase Readability

The name of the array is a constant pointer which contains the memory address of the _____ of the array”

First element

What is output of following code?

```
Int data[10]=[10,20,30,40,50,60,70,80,90,100];  
Int xptr=&data[4];  
Int*yptr=&data[6];  
Cout<<xptr+xptr++
```

120

C++ offers _____ levels of data access control inside a class

Three

The function calloc takes two arguments, first argument is the _ and the second argument is the _____

Required space in terms of numbers, size of space

The statement cout<<yptr will show _____ the yptr points to

Memory address

The use of *sign in first and second statements is called _____ and _____ of pointer.

```
Int * ptr = &x
```

```
Cout<<*ptr
```

Referencing dereferencing

What will be the output of following program?

```
Int data[10]=[10,20,30,40,50,60,70,80,90,100];
```

```
Int*yptr=&data[5];
```

```
Cout<<++yptr
```

61

Character string are terminated by _____

'\0'

There is a pointer variable named ptr of type float. Which type of variable address can be stored by ptr.

int

char **argv can be read as

array of pointers to char

eof is used to check for the _____ of file when a file is being

read end

suppose we have int y[10]; To access the 4th element of the array we write _____

y[4];
y[3];
y[2];
none of given

From the following; which one is the range of Random number generator function rand()?

0 – 32768
1 – 32768
0 – 32767
1 – 32767

From following; which one is the correct syntax of an array initialize: Array size is 10 and it is of double data type to value 0?

arr[10] = {0.0};
double arr[10]= 0.0;

```
double arr[10] = {0.0};  
double arr[] = 0.0;
```

The string in the array is terminated by a _____

zero
nil
null
one

In C/C++; by default arguments are passed by _____ to a function.

Reference
Value
Type Data

char name [] = "Hello World" ; In the above statement, a memory of _____ characters will be allocated

12

_____ Keyword is used to return some value from a function.

break
return

Which of the function call is call by value for the following function prototype? float add(int);

add(x);

The increment of a pointer depends on its _____.

data type

The ASCII code of null character is_____

000

Pointers are a special type of_____in which a memory address is stored
variables

Transpose of a matrix means that when we interchange rows
and columns_____

the first row becomes the Last column
the first row becomes the first column

Pointers store the _____
value of a variable

Which of the following function call is “call by reference” for the following function prototype? int add (int *);

add(&x);

The name of the array is a constant pointer which contains the memory address of the_of
the array.

first element

array name always contains the memory address of the _____ of the array

first element

Which function is used to delete the allocated memory space? **Free**

Symbolic constant PI can be defined

as **#define PI3, 14**

Windows operating system may itself takes memory from _____.

Heap

All the preprocessors directive starts with _____.#

Which one of the following is used to perform bit-wise OR operator

I

Constructor is itself a _____ of C++ and _____.

Function, can be overloaded

RAID stands for _____.

Redundant array of inexpensive devices

Object code is machine code but it is not _____ and _____.

Reloadable, executable

At the _____, we try to break up the problem into functional units analysis phase
design phase

For binary member operators, operands on the _____
drives (calls) the operation.

Left

We cannot increment _____.

references

We can _____ pointer.

all of the given

We can _____ references.

None of the given

What will be the correct syntax for the following function call?

float add (int &);

add(x);

An instance of a class is called_____.

object

The_____is called automatically when an object destroys

destructor

The destructor is used to_____.

1. deallocate memory

___data isn't accessible by non-member functions or outside classes.

private

Member functions of the class_____main program.

are accessible from

Overloading means :

Using the same name to perform multiple tasks or different tasks depending on the situation.

The main advantage of function overloading is

_____.

The program becomes more readable

You cannot overload the _____ operator.

1. ? :

In C++, a variable can be declared anywhere in the program this will increase _____.

efficiency

Memory allocated from heap or free store _____.

cannot be returned back unless freed explicitly using free and delete operators

We cannot use _____ pointer for storing and reading data from it.

'NULL

The dynamic memory allocation uses _____ whereas static memory allocation uses _____.

heap , stack

What will be the output of the given code?

```
#include #define MAX( A, B ) ((A) > (B) ? (A) : (B)) void main() {  
int i, x, y; x =  
23;  
y = 45;
```

```
i = MAX( x++, y++ );  
// Side-effect: // larger value incremented twice cout<< "x = " << x << " y = "  
<< y << '\n';  
}
```

x=24 y=47

NULL has been defined in _____ header file.

Stdlib.h

Symbolic constant PI can be defined as:

#define PI 3.14

The friend function of a class can have access

_____.

to the private data members

C++ was developed by _____.

2. BejarneStroustrup

Once the _____ are created, they exist for the life time of the program.

3. static variables

Encapsulation means _____.

4. that the data of a class can be accessed from outside

An address is a _____, while a pointer is a _____.

5. variable, constant

The syntax of declaration of a function that returns the reference to an integer is ____.

`int &myfunc();`

Which one of the following is mandatory preprocessor directive for c++?

`#include`

The members of a class declared with the keyword struct are ____ by default.

`public`

getche() is a _____ function and defined in _____ header file.

`built-in function ,conio.h`

_____ operators are the ones that require two operands on both sides of the operator.

`Binary`

_____ will return the number of bytes reserved for a variable or data type.

`sizeof operator`

_____ are not available in C language.

`Library functions`

The members of a class declared without any keyword are_____by default.

private

For console input and output we use_____.

conio.h header file

The name of the destructor is the same as that of a class proceeding with a__.

~ sign

A reference cannot be NULL it has to point a data type.

True

A pointer is_____.

a variable for storing address

Constructor is a special function, called whenever we
_____.

instantiate an object of a class

Symbolic constant PI can be defined as:

#define PI 3.14

Object code is machine code but it is not_____and
_____.

relocatable, executable

The default visibility for the data members of the class is

private

The__is called automatically when an object destroys.

destructor

Constructor is special type of function :

which has no return type

_____variables are those that are defined outside of main.

Global

Within the statement obj1=obj2; obj1 will call the assignment operator function and obj2 will be passed as an argument to function.

True

by Attia Kundi

When the compiler overload the assignment (=) operator by default then

Compiler does member wise assignment.

It is possible to return an object from function using this pointer.

True

Overloaded assignment operator must be

Member function of class

Let suppose int a, b, c,
d, e;

a = b = c = d = e = 42;

This can be interpreted by the compiler as

2. a = (b = (c = (d = (e = 42))));

In statement a+b+c, at first

b+c is executed first

Suppose int i = 10; then what is the output of cout<<oct<<i;

12

13

ostream is a _____ operator.

None of the given

_____ must be included to use stream manipulation in your code.

omanip

_____ operators are the ones that require only one operator to work.

Unary

The endl and flush are _____.

Manipulators

When operator function is implemented as member function then return

type of function_____.

When a variable is defined as static in a class then
_____.

Only one copy is created for all objects of this class

Automatic variables are created on_____.

Stack

```
cout<<i<< " "; cout<< d <<" ";  
cout<< f;
```

Above statements can be written within statement of one line as:

```
cout<<i<<" "<< d <<" "<< f;
```

dec, hex, oct are all_____.

Non-parameterized manipulators

What will be the output of following statement?
`cout<<setfill('0')<<setw(7)<< 128;`

2.0000128

Which of the following syntax is best used to delete an array of 5 objects named 'string' allocated using new operator.

`delete []string;`

If we have a program that writes the output data(numbers) to the disc, and if we collect the output data and write it on the disc in one write operation instead of writing the numbers one by one.

In the above situation the area where we will gather the number is called

Buffer

The first parameter of operator function for << operator

_____.

Must be passed by reference

The second parameter of operator function for >> operator must always be passed

By reference

The only operator that the compiler overloads for user define data type by default is

Assignment (=) operator

Consider the following code, the printed value will be converted into:

```
int n=10; cout<<oct<<n;
```

Base 8

_____ variables are defined in the main.

Local

ostream class is _____ and not under our control.

built-in

The memory allocation in C++ is carried out with the help of_____.

new operator

If B is designated as friend of A, B can access A's non- public members.

A can access non-public members of B

If the request of new operator is not fulfilled due to insufficient memory in the heap_____.

the operator returns 0

We should not use such variable names that are starting with `__` because in C++, there are lots of internal constants and symbolic names that start with it.

double underscore

The friend keyword provides access_____.

in one direction only

The malloc function takes_____argument(s).

one

The constructor contains_____.

no return type

What will be the output of the following c++ code? `#include<iostream.h>`

```
#define max 100 main()
{
#ifdef max
    Cout<<"Hellow;
}
```

Error

Once we have defined a symbolic constant value using `#define`, that value_____during program execution

cannot be changed

The memory allocation functions return a chunk of memory with a pointer of type_____.

ptr

A class can be declared as a_____of other class.

member

To avoid dangling reference, don't return_____.

the reference of a local variable from the function

Constructor is itself a_____of C++ and_____.

function, can be overloaded

The parameter passed to isdigit() function is _____ variable.

Integer

char **argv can be read as _____.

pointer to pointer to char

To read command-line arguments, the main() function itself must be given _____ arguments.

2

How many bytes an integer type pointer intPtr will jump in memory if the statement below is executed?

```
intPtr += 2 ;
```

8

The increment of a pointer depends on its _____.

data type

The statement cout<<yptr will show the _____ theyptr points to.

memory address

_____ is used as a dereferencing operator.

*

Transpose of a matrix means that when we interchange rows and columns_____.

the first row becomes the first column

Individual characters in a string stored in an array can be accessed directly using array_____.

subscript

We can define a matrix as _____ array.

Multi dimensional

A _____ is an array of characters that can store number of character specified.

String

Given a two dimensional array of integers, what would be the correct way of assigning the value 6 to the element at third row and fourth column?

`array[2][3] = 6 ;`

_____ of a variable means the locations within a program from where it can be accessed.

Visibility

Which of the following function call is "call by reference" for the following function prototype? `int add (int *)`;

`add(&x)`;

Which of the following function call is "call by reference" for the following function prototype? `float add (float *)`;

`add(&x)`;

Which of the function call is call by value for the following function prototype?

`float add(float)`;

`add(x)`;

Which of the function call is "call by value" for the following function prototype?

`float add(int)`;

`add(x)`;

Return type of a function that does not return any value must be_.

`void`

_____ will be used for enclosing function statements into a block.

`{ }`

What is the output of the following code if the 2nd case is true

```
switch (var) {  
  case 'a': cout<<"apple"<<endl; case  
  'b':cout<<"banana"<<endl; case  
  'm':cout<<"mango"<<endl; default: cout<<"any  
  fruit"<<endl;  
}
```

`banana`

`mango any`

fruit

When the break statement is encountered in a loop's body, it transfers the control from the current loop.

Outside

What is the output of the following code if the 3rd case is true

```
switch (var) {  
case 'a':cout<<"apple"<<endl; case  
'b':cout<<"banana"<<endl; case  
'm':cout<<"mango"<<endl; default: cout<<"any  
fruit"<<endl;  
}
```

mango

any fruit

What is the output of the following code, if the first case is true

```
switch (var) {  
case 'a':cout<<"apple"<<endl; case  
'b':cout<<"banana"<<endl; case  
'm':cout<<"mango"<<endl; default: cout<<"any  
fruit"<<endl;  
}
```

apple

banana

mango

any fruit

What will be the output of following code segment?

```
for (int i = 2; i<10; i++) { if ( i == 5)  
continue; cout<<i<<" ";  
}
```

2,3,4,6,7,8,9

__statement is used to terminate the processing of a particular case and exit from switch structure.

break

What will be the result of the expression $j = i++$; if initially $j = 0$ and $i = 5$?

5

What will be the result of the expression $k = ++m$; if initially $k = 0$ and $m = 4$?

5

What will be the result of the expression $k = ++m$; if initially $k = 0$ and $m = 5$?

6

How many times the following do-while loop will execute?

```
int k = 10; do { cout<< "Statements" <<endl; k -= 2; } while(k>0);
```

6

7

Which of the following loops checks the test condition at the end of the loop?

Do-While

The operators ++ and -- are used to increment or decrement the value of a variable by _____.

1

How many times the following loop will execute?

```
int j = 3; while(j > 0) { cout<< "Statements" <<endl; j -= 2; }
```

3.2

A__structure specifies that an action is to be repeated while some condition remains true.

Control

!(x> 3) means in C++ that

x is less than or equal to 3

When the logical operator && combines two expressions then the result will be true only when the both expressions are_.

true

< and > both are_____operators.

Relational

What will be the value of variable “input” if the initial value of input is 67?

```
if(input >= 50) input = input  
+ 1; if(input <= 75) input =  
input + 2; else  
input = input - 1;
```

3.70

!(x< 3) means in C++ that

x is greater than or equal to 3

!= operator is used to check whether the operand on the left-hand-side is_____to the operand on the right- hand-side.

Not equal

When the if statement consists more than one statement then enclosing these statement in curly braces is,

Must

The most suitable data type for number 325.25 is

float

What will be the result of arithmetic expression $6 + 48 / 4 * 3$?

42

Which of the following will be the most appropriate data type to store the value 63.547?

Float

In the given expression which operator will be evaluated first? $10 + (6 / 2) - 2 * 3$?

3. /

What will be the value of the variable output in the given piece of code?

```
double output = 0;  
output = (2 + 2) * 4 + 2 / (4 - 2);
```

17

It is the job of _____ to transfer the executable code from hard disk to main memory.

1. interpreter
2. Debugger
3. Linker
4. Loader

In computer systems there are mainly _____ type of softwares.

2

_____ will explain the function of a program.

Comments

if ($a > b$ && $a > c$) then the condition will be true only if

Both $a > b$ and $a > c$ are true

A variable of character data type occupies _____ byte(s) in memory.

1

We must include the header file _____ to convert the value of one type into another type using built-in functions.

string.h

A function is a block of statements that can be defined once and used _____ in the program.

As many times as user wants

Select the correct way to assign the address of first element of array to pointer?

int *ptr = &data;

Consider the following code segment. What will be the output of following code?

```
int addValue (int *a){ int b = (*a)
+ 2; return b;
}
main() { int x
= 6;
cout<<addValue(&x)<<" "; cout<<x;
}
```

6,8,6

Here the code is given below. You have to identify the problem in the code.

```
while(i< 10) && (i> 24))
```

the test condition is always false

The correct syntax of do-while loop is _____.

do {statements;} while (condition);

Matrix is defined as _____.

Multi-dimensional array

In programming, comments are used to explain the functioning of the_.

Program

by Attia Kundi

Operating System is a type of a/an_____.

system software

From the options given, you need to choose the option which is true for the given code.

```
for (int i = 1; i>0; i++) {  
    /*loop code*/  
}
```

the test condition is always true

Which of the following values are used in C/C++ to represent true and false?

1 and 0

'While' loop may execute_____or more times.

zero

Body of any function is enclosed within_____.

1. { }

What will be the correct syntax for initialization of a pointer ptr with string "programming"?

char *ptr = "programming";

Which one of the given option is not a mode for reading/writing the data from a file?

4. get

Which of the following operators is used to access the value of variable pointed by a pointer?

* operator

In case of single dereferencing, the value of the

_____is the address of the_____.

pointer, variable

The remainder (%) operator is a_____operator.

Arithmetic

CS201

Mid-Term

Quiz by Attiq Kundi

SUBSCRIBE "Knowledge For Us" ON YOU TUBE

100 % CORRECT MCSQ

AND Many MoRe

CS 201

What will be the output of following code? `int x = 10;`

```
cout<<"x="<<x;
```

x=10

The purpose of using `cout<<` is to_____.

Display information on the screen

Which of the following data types will be assumed if no data type is specified with constant?

short

When an array element is passed to a function, it is passed by_____.

reference

While programming, it is good to provide an easy to understand and easy to use interface; this programming skill is called_____.

usability

_____executes all the lines before error and stops at the line which contains the error.

Intrepreter

Which of the following is the correct syntax to access the value of first element of an array using pointer ptr?

`*ptr[0]`

C is a/an _____ language.

`function oriented`

_____ of a function is also known as signature of a function.

`Declaration`

_____ are very good tools for code reuse.

`functions`

If any break statement is missed in switch statement then_.

`this may cause a logical error`

A 2D array `multi[5][10]` can be accessed using the array name as `**multi`, this technique is called_____.

`Double dereferencing`

In C/C++, the default command line arguments passed to the main function are_.

`int argc, char **argv`

A record is a group of related_____.

`Fields`

The microsoft word document (.doc) is a kind of_____.

`Random AccessFile`

NULL character is used to indicate the_____ of string.

`End`

How many dimensions does n-dimensional array has?

`n dimensions`

Which of the following function call is "call by reference" for the following function prototype?

`func(&num);`

The loop which is most suitable to be used when the number of iterations is known is called_____.

`for`

In C/C++, the string constant is enclosed in_____.

double quotes " "

In order to get the right most digit of a number, we divide this number by 10 and take_____.

Its remainder

What is the correct syntax to declare an array of size 10 of int data type? ●

int name[10];

How many bytes of memory are occupied by array 'str'? char str[] = "

11

Suppose that an integer type pointer contains a memory address 0x22f230. What will be the new memory address if we increment this pointer by one?

0x22f234

Which of the following if missing would result in infinite recursion in case of recursive function?

Recursive call

Whenever we use a library function or a predefined object or macro, we need to use a_____.

header file

Switch statement deals with_____type of data.

Both Integer and Character

Both compiler and_____are used to translate program into machine language code.

interpreter

TWAIN stands for_____.

Technology Without An Interesting Name

The parameter passed to isdigit() function is_____.

a character variable

C++ views each file as a sequential stream of_____.
_____.

Bytes

Structure is a collection of_____under a single name.

only variables

The default mode for writing into a file using ofstream object is _.

out

The memory address of the first element of an array is called ____.

base address

We want to access array in random order which of the following approach is better?

Pointer

The __ structure is a multiple-selection construct which makes the code more efficient and easy to read and understand.

switch

Which of the following is not a reserved word in C/C++?

sum

To access rand(), which library is required to be included in program?

stdlib.h

What is the highest legal index for the following array? int arr[4]

4

Word processor is a type of a/an _____.

application software

Identify the correct option which is used for calling the function float area (int).

area(int num);

The _ statement allows us to select from multiple choices based on a set of fixed values for a given expression.

switch

C is widely known as development language of _____ operating system.

Unix

To convert the value of one type into another type using built-in functions, we include _____ header file.

stdlib.h

The keyword _____ is used to get some value back from a function.

return

The function seekg() takes _____ parameter(s).

2

The function write() takes _____ as parameter(s).

String and no. of bytes to be written

When the logical operator AND (&&) combines two expressions exp1 and exp2 then the result will be true only

_____.

When both exp1 and exp2 are true

Syntax of a union is identical to _____.

structure

Which one of the symbol is used to represent a decision in a flow chart?

Rectangle

In Flow Chart, flow of control is represented by

_____.

4. Arrow

There can be _____ 'default' statement(s) in any switch structure.

2

The condition in loop should be a(n) _____.

Arithmetic Expression

How many nested loops would be required to manipulate n-dimensional array?

2n

Which of the following is not an example of int data type?

-4

We should use _____ for clarity and to force the order of evaluation in an expression.

parenthesis ()

Which of the following is the starting index of an array in C++?

0

The statement $x += y$ can be interpreted as_____.

Adding the value of the x to the value of the y and storing the result in x

Given a 2D array of integers, what would be the correct way of assigning the value 5 to the element at second row and third column?

$m[1][2] = 5;$

Array is a data structure that stores_____.

Variables

A program statement that invokes a function is called
_____.

function call

If a function has been declared but not defined before its function call then it is termed as_____.

run time error

The compiler of C language is written in _____ language.

C

Which one of the below functions is not included in ctype.h header file?

getdigit(int c)

Which function is used to locate the first occurrence of a character in any string?

strlen()

To access the data members of structure, _____ is used.

Dot operator

In the following nested For Loop, which loop will run most number of times?

```
for( inti = 0; i < 5; i++)  
{
```

```
    for(int k = 0; k < 5; k++)  
    {  
        .....  
    }  
}
```

Depends upon the statements in the inner loop's body

Structure use _____ allocation.

_____ function give the position of the next character to be read from that file.

tellp()

What will be the size of the following character array? char name[] = "Adeel";

5

Function prototype is written,

Before call of that function

Which one of the following languages has been used to write the compiler of "C" language?

C

A hierarchy of classes which are used to deal with console and disk files are called_____.

Stream classes

____stops execution at the line that contains error(s) in the code.

Interpreter

C++ is a_____language.

High level

How many elements are stored in the following? int matrix [4][5];

20

____is a substitute of multiple if statement.

if. . .elseif statement

if

int sum = 54;

Then the value of the following statement is sum = sum - 3;

3. 51

What will be the correct syntax for declaration of the following statement?

"ptr is a constant pointer to an integer"

const int *myptr;

____operator is used to pass the address of a variable in call by reference method.

&

____data type can operate on modulus operator.

int

Whenever some number is added in an array name, it will jump as many_ as the added number.

rows

Suppose that an integer type pointer contains a memory address 0x22f220. What will be the new memory address if we increment this pointer by one?

0x22f224

___ is the pointer which determines the position in a file from where the next read operation occurs.

put

Destructor _____

Takes no arguments and Can be overloaded

The default visibility for the data members of the class is _____.

Private

Which of the following permits function overloading in C++ Both

type and arguments

Once we have defined a symbolic constant value using #define, that value _____ during program execution.

Cannot be changed

Encapsulation means _____.

That the data of a class cannot be accessed from outside

The original value remains unchanged in _____

Call by value

Programmer should be very careful about the memory management because it can _____.

cause problems of memory leakage and dangling pointers

The destructor is used to _____.

deallocate memory

The main advantage of macros is that _____ and program runs faster
macros are more efficient

Constructor is special type of function :
which has no return type

Unix operating system was written in C language
C

When accessing a structure member, the identifier to the left of the dot operator is the name of _____.
Structure variable

Constructor has _____.
the same name as of class

For console input and output we use _____.
conio headerfile

We should not use such variable names that are starting with _____ because in C++, there are lots of internal constants and symbolic names that start with it.

upper case alphabets

CS201

Mid-Term

Quiz by Attiq Kundi

SUBSCRIBE "Knowledge For Us" ON YOU TUBE

100 % CORRECT MCSQ
AND Many MoRe

CS 201

by Attia Kundi